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Large deformation analysis of granular materials with stabilized and noise-free stress treatment in smoothed particle hydrodynamics (SPH)

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ABSTRACT

This paper presents a new smoothed particle hydrodynamics (SPH) model for the large deformation analysis of granular materials with improved accuracy and stability. In order to remove the spurious stress profile during the post-failure process which is a common issue for the application of SPH to geotechnical problems, an innovative stress diffusion term is formulated. Further a new boundary treatment is proposed with the use of renormalization techniques to give a first-order consistent wall boundary conditions for geotechnical applications. This methodology is able to deal with complex geometries such as sharp corners. A number of test cases are considered to examine the robustness and accuracy of the proposed technique. Results show that the new boundary formulation is able to improve the accuracy of the solution even for a relatively coarse particle resolution. The stress diffusion term reduces the numerical noise that affects the stress under large deformations, resulting in a smooth stress field. The proposed SPH model is finally applied to the simulation of granular flow and tunnel face collapse, showing satisfactory agreement and convergence with experimental results.

1. Introduction

The geophysical flows involving large deformations are closely related to a number of natural phenomena and engineering applications, including snow and rock avalanches, debris flow in disaster prevention, landslides in slope stability, soil liquefaction in seismic design, and internal erosion in dam maintenance. Numerical analysis is a powerful tool to study the progressive failure in geophysical flows and predict their potential catastrophic consequences, whereas the presences of free surfaces and extremely large deformation are the main challenges of the modelling of such flows (Huang and Dai 2014).

In computational geomechanics, the most widely used predictive tools are based on the finite element (FE) and finite difference (FD) methods. These methods are very efficient in solving partial differential equations (PDEs), and have been widely applied in the studies of geotechnical problems. Despite the success of mesh-based methods, the use of meshes still presents some numerical difficulties, such as mesh distortion and domain discretisation, which hinder their applicability to problems with large deformation and free surfaces, such as those encountered in granular flows and post-failure analysis of particulate materials. The mesh regeneration technique (Wang et al. 2013) provides a possible solution for solving the mesh distortion problems in FE and FD methods, however this is computationally demanding and not always practical.

Comparing to mesh-based methods, a more efficient solution for the large deformation problem is provided by meshless methods in which the mesh connectivity is eliminated. Examples of meshless methods used for granular materials and geotechnical applications include Smooth Particle Hydrodynamics (SPH) (Gingold and Monaghan, 1977), the Material Point Method (MPM) (Sulsky et al., 1994), and the Particle Finite Element Method (PFEM) (Onate et al., 2004). Among them, SPH can be regarded as a truly meshless, Lagrangian method. In SPH, the physical domain is discretized into a set of Lagrangian particles carrying macroscopic field properties, and moving with the material velocity. The interaction between particles is controlled by a weighting function called a smoothing kernel. A background mesh is not needed for the integration of the governing partial differential equations since the local interpolation is adopted to evaluate the field variables of each particle. These features make SPH ideal for the problems involving high nonlinearity, a free surface and large deformations.

Although SPH was originally developed for astrophysical problems by Gingold and Monaghan (1977), and Lucy (1977), in the past decades it has been applied to a wide range of applications, including fluid dynamics (Shao and Lo, 2003), fracture of solids (Benz and Asphaug,

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1995), multi-phase flow (Colagrossi and Landrini, 2003), soil–water interaction (Bui et al., 2007), and debris flow (Pastor et al., 2009). The application of SPH in computational geomechanics was pioneered by Bui et al. (2008). In their work, the elastic-perfectly plastic Drucker-Prager model with associated and non-associated plastic flow rule was successfully implemented in the SPH scheme for the study of the post-failure flow of cohesive and non-cohesive soils. Later, various rigorous constitutive models, such as Bingham flow model, rate-dependent viscoplastic model, hypoplastic model, critical-state based constitutive model, and $\mu(I)$ rheological constitutive model, were incorporated into the SPH framework for the modelling of different flow behaviour (Huang et al. 2012, Prime et al., 2014, Peng et al., 2015, Yin et al., 2018, Yang et al., 2020). Furthermore, there are also several studies devoted to eliminate shortcomings such as tensile instability within SPH for large deformation geomechanics problems, in which they combined SPH with other methodologies to as in Taylor-Galerkin SPH (Blanc and Pastor, 2013), and Stress-Particle SPH (Chalk et al., 2020).

In most existing geotechnical applications of SPH, the focus was primarily on the prediction of kinematics and satisfactory results were obtained in comparison with experimental observations. However, when analysing the stress distribution, the high-frequency short-length oscillations are observed under the large deformation. Similar to the pressure fluctuations observed in weakly compressible SPH for fluids (Molteni and Colagrossi 2009), this spurious stress field affects the accuracy and reduces the predictive capability of the SPH method since an accurate prediction of stress is very important for some practical applications (such as the simulation of impact of debris flow on structure).

The stress regularisation technique recently proposed by Nguyen et al. (2017) provided a solution for the spurious stress field by performing a filter over the entire domain to re-evaluate and re-assign the stress of each particles in a certain time frequency. Nguyen et al. (2017) suggested the regularisation of stress with moving least squares (MLS) interpolants to achieve first-order correction. In their work, the regularisation technique successfully filtered the high-frequency numerical noise in the stress field. However, a common issue associated with using a filtering technique in SPH is that the field variables may be over-filtered for long-time simulations, thereby affecting the physics involved in the process being simulated (e.g., Sibilla, 2007). In addition, the filtering procedure itself requires an extra particle sweep over the entire domain, in which the computational costs are expensive, especially for a large dimension problem. Therefore, it is necessary to find an alternative for stress smoothing with the ability to maintain the momentum conservation and consistency.

Another important issue associated with SPH is the treatment of the boundary conditions (Vacondio et al. 2020). The accuracy and robustness of numerical simulation is highly dependent on the performance of boundary conditions adopted in SPH models. For the application of SPH in geomechanics, the most widely used boundary treatment is the one proposed by Bui et al. (2008). In their work, for a soil particle near the boundary, the stress values of the neighbouring boundary particles were assigned to be identical with that of this soil particle, giving a locally uniform stress profile near the solid boundary. The no-slip boundary condition was imposed by constructing an artificial velocity for boundary particles as in Morris et al. (1997). Several corrections for Bui's boundary condition were proposed in the later studies. Peng et al. (2015) argued that Bui's treatment on the stress of boundary particles may fail to guarantee a no-penetration condition and proposed to take the diagonal components of stress tensor of the boundary particles as the maximum value of the soil particles of concern, while the off-diagonal components were set to be the same; this ensured a sufficiently large pressure generated from the boundary. A different modification for the same issue was proposed by Chalk et al. (2020), who employed a single-layer of dummy wall particles that exert repulsive forces to ensure the no-penetration condition. Another possible boundary treatment relies on the use of dummy particles and renormalization technique (e.g., Peng et al., 2019; Yang et al., 2020). The field quantities (e.g., stress, velocity) of dummy boundary particles are extrapolated from their neighbouring soil particles by a Shepard corrected kernel function to give a zero-th order consistency. It is also worth mentioning the work by Douillet-Grellier et al. (2017) and Zhao et al. (2019) on the development of stress boundary condition with no boundary particles. The basic idea of their treatment is to complete the truncated kernel support of the particles near the boundary with an extra stress term included in the momentum equation. This type of Dirichlet wall boundary condition can be applied when boundary stresses are required, such as in the simulation of triaxial test, Brazilian test, and penny-shaped crack problem.

Available boundary treatment techniques for geotechnical application of SPH can ensure only up to zero-th order consistency. The lack of consistency in boundary treatment may result in poor prediction of the stress field close to the boundaries. In addition, although many of these boundary methods work well for test cases with simple geometries, their performance has not been assessed for more complex geometries.

This work develops a new method that can minimise the spurious stress field commonly involved in SPH modelling of geomechanical problems and address to some of the limitations of existing boundary conditions. Inspired by density diffusion scheme in fluid dynamic (Antuono et al., 2012; Jandaghian and Shakibaeinia, 2020), which gives a consistent numerical algorithm for removing the pressure noise and improving stability, the diffusion process is considered in this study to improve the stress field. In order to accurately resolve the stress profile near the boundary while being able to handle complex geometries, the fixed ghost particle technique proposed by Marrone et al. (2011) is modified in this work to treat the wall boundary condition. To the best of authors' knowledge, it is the first time for such a boundary technique extended to model the elasto-plastic solid with stress in SPH. Instead of using MLS interpolant to evaluate the flow quantities as presented by Marrone et al. (2011), the corrected SPH interpolation proposed by Liu and Liu (2006) is employed to restore the first-order consistency at the boundary. This procedure can be achieved within a single loop, thereby minimising the computational costs.

The remaining part of this paper is organised as follows: Section 2 presents a brief description of the governing equations and constitutive model. Subsequently, the SPH discretisation of governing equations is given in Section 3 together with the development of a numerical diffusion term. Section 4 provides a detailed description of the first-order consistent wall boundary treatment. In Section 5, the accuracy and robustness of the new SPH scheme is assessed with several validation cases, including a Couette flow, simple shear test, static soil with wedge, 2-D granular column collapse, and tunnel face collapse. Finally, the main findings and conclusions are presented in Section 6.

2. Numerical model

2.1. Governing equations

In continuum mechanics, the boundary-valued problem needs to be formulated as a set of differential equations with initial/boundary conditions. The conservation of mass and momentum for a particulate material with density ρ yields the following differential equations,

$$\frac{D\rho}{Dt} = -\rho\nabla\cdot\mathbf{v} \tag{1}$$

$$\frac{D\mathbf{v}}{Dt} = \frac{1}{\rho}\nabla\boldsymbol{\sigma} + \mathbf{f} \tag{2}$$

where $\mathbf{v}$ is the velocity, $\boldsymbol{\sigma}$ is the total stress tensor (which is negative for compression in this study), and $\mathbf{f}$ is the external body force.

The above equations are independent of the mechanical properties of the particulate material and are not sufficient to complete the system of equations required to solve the boundary value problem. An additional constitutive equation is therefore needed to relate stress and strain as discussed in further detail in the following section.

2.2. Constitutive model

In continuum mechanics, the stress and strain are tensorial quantities whose relationship is expressed in terms of constitutive equations. In the present work, the equations are presented using the indicial notation and Einstein convention, where α, β and γ denote the Cartesian coordinates.

Elasto-plastic constitutive models have been extensively used to capture the real behaviour of soils and to simulate large deformation problems of geomaterials (e.g., Bui et al., 2008; Chen and Qiu, 2012; Nguyen et al., 2017). The elasto-plastic constitutive relation is defined in terms of a yield function and a plastic potential, or flow-rule. The yield function f separates the purely elastic response from the elasto-plastic behaviour. The plastic potential function g determines the direction of plastic strain increment.

The present work adopts the Drucker-Prager yield criterion with non-associated flow rule, whose yield function and plastic potential function are given by,

$$f = \alpha_\phi I_1 + \sqrt{J_2} - k_c \tag{3}$$

$$g = \alpha_\psi I_1 + \sqrt{J_2} \tag{4}$$

where I_1 is the first principal stress invariant, J_2 is the second deviatoric stress; α_ϕ and k_c are Drucker-Prager constants. Their values can be related to Coulomb's material constants c (cohesion) and ϕ (internal friction). For plane strain condition, the relation between Drucker-Prager constants and material constants is given by,

$$\alpha_\phi = \frac{tan\phi}{\sqrt{9+12tan^2\phi}} \tag{5}$$

$$k_c = \frac{3c}{\sqrt{9+12tan^2\phi}} \tag{6}$$

α_ψ is a dilatancy factor. Its value can be related to the dilation angle ψ in a similar manner to that between α_ϕ and friction angle φ (Bui et al., 2014, Nguyen et al., 2017).

Substituting the yield function and plastic potential function, i.e., Eqs. (3) and (4), into the generalized form of the elastic-perfectly plastic model shown in Appendix A, i.e, Eq. (A11), the stress–strain relationship for Drucker-Prager elastic-perfectly plastic constitutive model can be expressed as,

$$\dot{\sigma}^{\alpha\beta} = 2G\dot{e}^{\alpha\beta} + K\dot{\varepsilon}^{\gamma\gamma}\delta^{\alpha\beta} - \dot{\lambda}\left[3K\alpha_\psi\delta^{\alpha\beta} + \frac{G}{\sqrt{J_2}}s^{\alpha\beta}\right] \tag{7}$$

where $\dot{\sigma}^{\alpha\beta}$ is the stress rate tensor, $\dot{\varepsilon}^{\alpha\beta}$ is the strain rate tensor, $\dot{e}^{\alpha\beta}$ is the deviatoric strain rate tensor, G and K are the elastic shear modulus and elastic bulk modulus, respectively, $\delta^{\alpha\beta}$ is the Kronecker's delta, $s^{\alpha\beta}$ is the deviatoric stress, and $\dot{\lambda}$ is the rate of plastic multiplier that is given by

$$\dot{\lambda} = \frac{3\alpha_\phi K\dot{\varepsilon}^{\gamma\gamma} + \left(G/\sqrt{J_2}\right)s^{\alpha\beta}\dot{\varepsilon}^{\alpha\beta}}{9\alpha_\phi K\alpha_\psi + G} \tag{8}$$

For large deformation problems, the Jaumann state that is invariant to rigid-body rotation is introduced. The final form of the stress–strain relationship is,

$$\dot{\sigma}^{\alpha\beta} = \sigma^{\alpha\gamma}\dot{\omega}^{\beta\gamma} + \sigma^{\gamma\beta}\dot{\omega}^{\alpha\gamma} + 2G\dot{e}^{\alpha\beta} + K\dot{\varepsilon}^{\gamma\gamma}\delta^{\alpha\beta} - \dot{\lambda}\left[3K\alpha_\psi\delta^{\alpha\beta} + \frac{G}{\sqrt{J_2}}s^{\alpha\beta}\right] \tag{9}$$

where the strain rate tensor $\dot{\varepsilon}^{\alpha\beta}$ and the spin rate tensor $\dot{\omega}^{\alpha\beta}$ can be related to velocity gradient from kinematic relations according to

$$\dot{\varepsilon}^{\alpha\beta} = \frac{1}{2}\left(\frac{\partial v^\alpha}{\partial x^\beta} + \frac{\partial v^\beta}{\partial x^\alpha}\right) \tag{10}$$

$$\dot{\omega}^{\alpha\beta} = \frac{1}{2}\left(\frac{\partial v^\alpha}{\partial x^\beta} - \frac{\partial v^\beta}{\partial x^\alpha}\right) \tag{11}$$

The two-step elastic predictor-plastic corrector scheme, also known as return mapping algorithms is adopted for the integration of the constitutive model. This procedure involves first calculating an elastic trial solution by integrating the elastic constitutive equations with strain increment. The predicted stress is then examined against the yield function. If the stress lies within or on the yield surface, the trial solution is accepted, otherwise, the plastic corrector step is performed to return the trial stress to the yield surface by correcting the plastic strain increment iteratively. More details of the algorithm can be found in Huang and Griffiths (2009).

3. SPH formulations

3.1. SPH formalism

In SPH, the integral approximation of spatial function $f(\mathbf{x})$ at the point $\mathbf{x}$ is defined as,

$$\langle f(\mathbf{x})\rangle = \int_\Omega f(\mathbf{x}')W(\mathbf{x}-\mathbf{x}',h)d\mathbf{x}' \tag{12}$$

where Ω is the interpolation region, W is the smoothing kernel function, h is the smoothing length that defines the extent of the support domain of the kernel function and $\langle\cdots\rangle$ denotes the SPH interpolation.

The discrete form of Eq. (12) can be written as,

$$\langle f(\mathbf{x})\rangle_i = \sum_{j=1}^{N} f_j W_{ij} V_j \tag{13}$$

where the subscript i and j stand for the interpolating particle and its neighbours, respectively, $W_{ij} = W(\mathbf{x}_i\text{-}\mathbf{x}_j, h)$ and $f_j = f(\mathbf{x}_j)$ are respectively the value of the kernel function W and the function f at the particle j with the position $\mathbf{x}_j$, N is the number of particles distributed in the support domain and V_j represents the associated volume of particle j.

Similar to the derivation of Eq. (13), the integral approximation of the gradient of a function $\nabla f(\mathbf{x})$ reads,

$$\langle \nabla f(\mathbf{x})\rangle = \int_\Omega \nabla f(\mathbf{x}')W(\mathbf{x}-\mathbf{x}',h)d\mathbf{x}' \tag{14}$$

The discrete approximation of Eq. (14) is given by,

$$\langle \nabla f(\mathbf{x})\rangle_i = \sum_{j=1}^{N} f_j \nabla_i W_{ij} V_j \tag{15}$$

where $\nabla_i W_{ij} = \nabla_i W(\mathbf{x}_i\text{-}\mathbf{x}_j, h)$ is the derivative of the kernel function with respect to particle i.

A variety of kernel functions can be found in literature. In this study, the 5th-order Wendland kernel C^2 is adopted. In studies by Robinson (2009) and Dehnen and Aly (2012), it is shown the Wendland function possess a non-negative Fourier transform indicating that it is stable against clumping instability. The kernel function takes the following form,

$$W(q,h) = \begin{cases} \alpha_d\left(1-\frac{q}{2}\right)^4(2q+1) & 0\leqslant q\leqslant 2 \\ 0 & q>2 \end{cases} \tag{16}$$

where α_d is the normalization constant. In one-, two- and three-dimensional space, α_d is equal to $3/4h$, $7/(4\pi h^2)$, and $21/(16\pi h^3)$ respectively and q is the dimensionless distance between points at $\mathbf{x}$ and $\mathbf{x}'$, defined as $q = |\mathbf{x} - \mathbf{x}'|/h$.

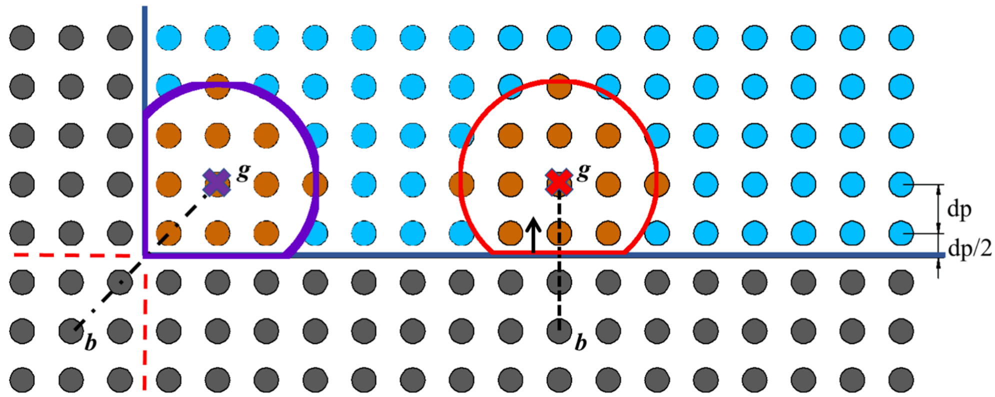

Fig. 1. Generation of boundary particles, *b*, (grey circles) and ghost nodes, *g*, (crosses) in flat surface and corner (soil particles are coloured by blue, those included in kernel summation of ghost nodes are coloured by orange). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

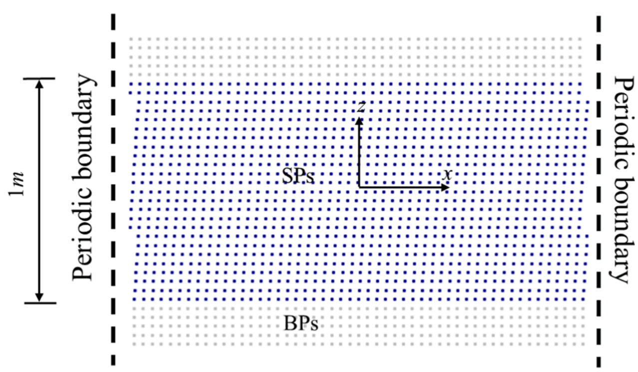

Fig. 2. Numerical set-up of the Couette flow case, in which soil particles (SPs) and boundary particles (BPs) are marked as dark blue and light grey.

3.2. SPH discretization of governing equations

With the use of standard SPH divergence and gradient operator, the governing equations (1) and (2) take following form,

$$\left\langle \frac{\mathrm{d}\rho}{\mathrm{d}t} \right\rangle_i = \rho_i \sum_{j=1}^{N} \frac{m_j}{\rho_j} \left(v_i^\alpha - v_j^\alpha \right) \frac{\partial W_{ij}}{\partial x_i^\alpha} \tag{17}$$

$$\left\langle \frac{\mathrm{d}v^\alpha}{\mathrm{d}t} \right\rangle_i = \frac{1}{\rho_i} \sum_{j=1}^{N} \frac{m_j}{\rho_j} \left(\sigma_i^{\alpha\beta} + \sigma_j^{\alpha\beta} \right) \frac{\partial W_{ij}}{\partial x_i^\beta} + g^\alpha \tag{18}$$

where, for a field function *f*, the subscript *ij* denotes the difference in value in the function $f_{ij} = f_i\text{-}f_j$ and *g* is the gravitational force. The gradient and divergence operators used in Eqs. (17) and (18) are skew-adjoint for the benefit of energy conservation (Mayrhofer et al. 2013).

It is well known that SPH exhibits unphysical oscillation and numerical instability if a numerical dissipative term is not included in the governing equations (Antuono et al. 2012). To dissipate these numerical fluctuations, an artificial viscosity term is required to be added in the momentum equation. This term was firstly designed to handle shocks (Monaghan, 1992) and has been found to stabilise the numerical algorithm. In this study, the artificial viscous term is introduced into the pressure term of the momentum equation in the following way,

$$\left\langle \frac{\mathrm{d}v^\alpha}{\mathrm{d}t} \right\rangle_i = \sum_{j=1}^{N} m_j \left(\frac{\sigma_i^{\alpha\beta} + \sigma_j^{\alpha\beta}}{\rho_i \rho_j} - \Pi_{ij} \delta^{\alpha\beta} \right) \frac{\partial W_{ij}}{\partial x_i^\beta} + g^\alpha \tag{19}$$

where

$$\prod_{ij} = \begin{cases} \dfrac{-\alpha_\prod \bar{c}_{ij} \mu_{ij} + \beta_\prod \mu_{ij}}{\bar{\rho}_{ij}} & v_{ij}^\alpha x_{ij}^\alpha < 0 \\ 0 & v_{ij}^\alpha x_{ij}^\alpha > 0 \end{cases} \tag{20a}$$

$$\mu_{ij} = \frac{h v_{ij}^\alpha x_{ij}^\alpha}{\left(x_{ij}^\alpha \right)^2 + \eta^2} \tag{20b}$$

$$\bar{c}_{ij} = 0.5(c_i + c_j) \tag{20c}$$

$$\bar{\rho}_{ij} = 0.5(\rho_i + \rho_j) \tag{20d}$$

with α_Π and β_Π are empirical constants used to control the magnitude of viscous dissipation and normally take values from 0 to 1. According to Monaghan (1992), the first term with α_Π produces a shear and bulk

Table 1
Adopted parameters in case study.

Parameters	Unit	Adopted values for test cases						
		Element shear tests		Static dry soil		Granular flow		Tunnel
		Couette flow	Simple shear test	Flat	Wedge	Case1	Case2	
h/dp	–	2.2	2.2	1.8	1.8	1.8	1.8	1.8
Diffusion coefficient	–	–	–	0.1	0.1	0.1	0.1	0.1
Viscous coefficient	–	1.0	1.0	0.1	0.1	0.1	0.1	0.1
Courant number	–	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Density	kg/m^3	2100	2100	2100	2100	1850	2079.5	2212.03
Elastic modulus	MPa	5.0	10.0	15.0	15.0	10.0	5.84	5.84
Possion ratio	–	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Friction angle	deg	–	30	25	0	22	21.9	21.9
Cohesion	kPa	–	20	0	–	0	0	0
Dilatancy angle	deg	–	0	0	–	0	0	0

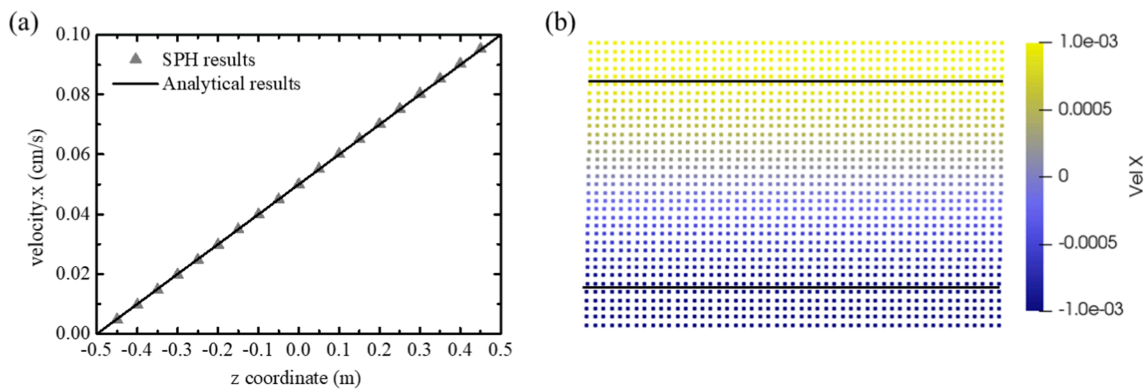

Fig. 3. Velocity profile of Couette flow: (a) comparison against analytical value and (b) velocity contour (m/s) at $t = 5$ *s*.

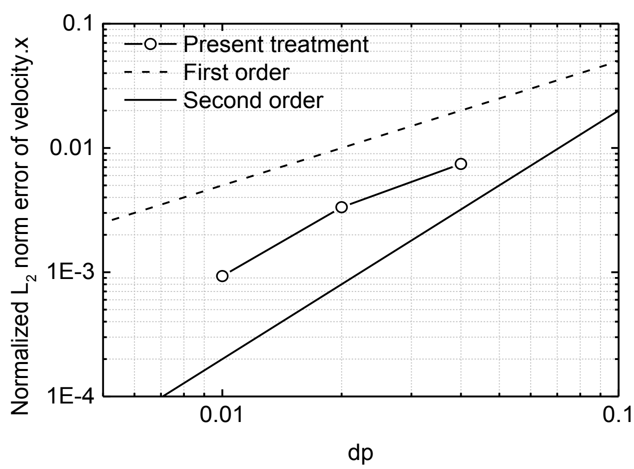

Fig. 4. Normalised L_2 error norms of the velocity at $t = 5$ *s* for Couette flow.

viscosity whereas the second term with β_{Π} is analogous to the Von Neumann-Richtmyer viscosity for high Mach number flow. For the modelling of geomaterials, the value of β_{Π} contributes little to the dissipation and thus can be omitted (Mao et al., 2017). The factor $\eta = 0.1\,h$ is inserted to the denominator to avoid numerical singularity. The speed of sound c of the material, is computed according to,

$$c_i = \sqrt{E_i/\rho_i} \tag{21}$$

where E is Young's modulus of the soil.

Furthermore, the equations for strain rate and spin rate of Equations (10) and (11) in SPH formulism are given by,

$$\langle \dot{\varepsilon}^{\alpha\beta} \rangle_i = \frac{1}{2}\left(\sum_{j=1}^{N} \frac{m_j}{\rho_j}\left(v_j^\alpha - v_i^\alpha\right)\frac{\partial W_{ij}}{\partial x_i^\beta} + \sum_{j=1}^{N} \frac{m_j}{\rho_j}\left(v_j^\beta - v_i^\beta\right)\frac{\partial W_{ij}}{\partial x_i^\alpha} \right) \tag{22}$$

$$\langle \dot{\omega}^{\alpha\beta} \rangle_i = \frac{1}{2}\left(\sum_{j=1}^{N} \frac{m_j}{\rho_j}\left(v_j^\alpha - v_i^\alpha\right)\frac{\partial W_{ij}}{\partial x_i^\beta} - \sum_{j=1}^{N} \frac{m_j}{\rho_j}\left(v_j^\beta - v_i^\beta\right)\frac{\partial W_{ij}}{\partial x_i^\alpha} \right) \tag{23}$$

3.3. *Stress diffusive term*

Although SPH can capture well the kinematics of geomaterial under large deformation, there are still some drawbacks on the prediction of the stress field due to the presence of high-frequency noise. This issue can be attributed to the collocated nature and lack of consistency of SPH. When the initial conditions are released in the early stages of simulation, numerical oscillations or fluctuations are appear which cannot be dissipated without an artificial dissipative term. Following the idea of density diffusive term, which has been shown to provide excellent results in fluid dynamics (e.g., Molteni and Colagrossi 2009, Antuono et al. 2012), a stress diffusion term is investigated in this study and included in the stress equation, for the purpose of giving a more accurate and smoother stress profile.

In order to achieve a smoothing process, according to Antuono et al. (2012), the diffusion term should approximate even derivatives of physical field, for example, a Laplacian operator, and it should vanish as the numerical accuracy increases to avoid unphysical behaviour and to recover the consistency (hence the diffusion term should be multiplied by smoothing length h to make its influence decrease when the spatial resolution increases). Meanwhile, the diffusion term should be defined to satisfy the global conservation of momentum, and global convergence.

Accordingly, the diffusion term proposed by Antuono et al. (2012) has a general form as follows,

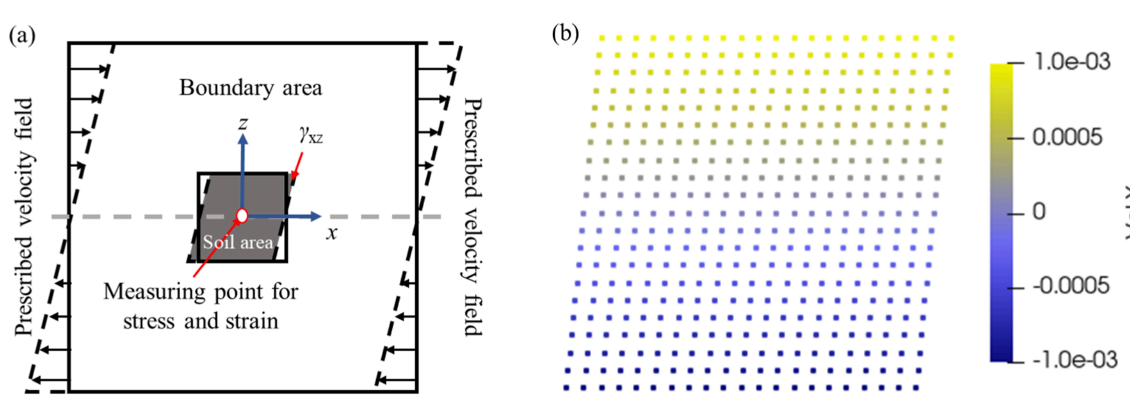

Fig. 5. Simple shear experiment for testing the Drucker-Prager elasto-plastic model in SPH (a) model set-up; (b) the obtained velocity profile.

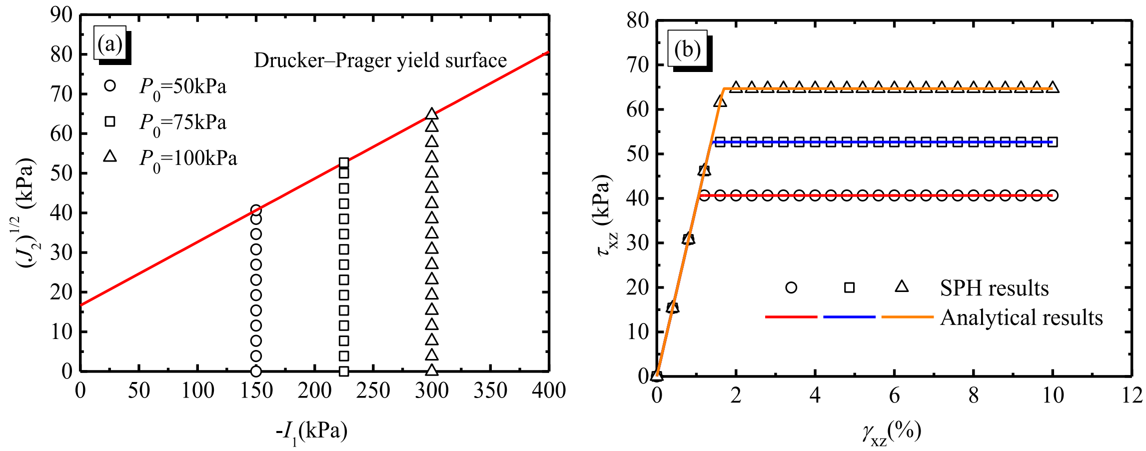

Fig. 6. Results of simple shear tests with varying confining stresses: (a) stress path and (b) shear stress–strain relationships.

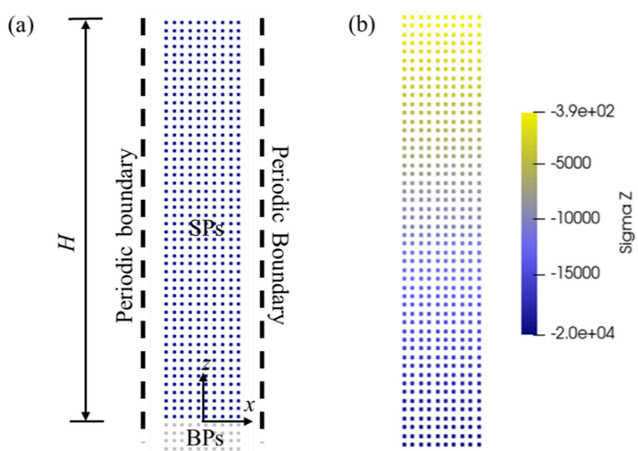

Fig. 7. Static soil column (a) model set-up, in which soil particles and boundary particles are marked as dark blue and light grey, respectively; (b) vertical stress contour. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$$D_i = 2\zeta h c_0 \sum_j \psi_{ji} \frac{\mathbf{x}_{ji} \cdot \nabla_i W_{ij}}{\mathbf{x}_{ji}^2 + \eta^2} \frac{m_j}{\rho_j} \tag{24}$$

where ζ is the coefficient used to control the magnitude of diffusion and normally take values as 0.1 for most application. The parameter ψ_{ji} is a diffusion operator and changes for different formulations. It has the dimension of the physical quantities to be diffused.

Under the hypotheses made above, the stress rate equation (9) should be written as,

$$\frac{D\sigma_i^{\alpha\beta}}{Dt} = D^{ep}\dot{\varepsilon}_i^{\alpha\beta} + D_i^{\alpha\beta} \tag{25}$$

where D^{ep} is the elasto-plastic stiffness matrix and $D_i^{\alpha\beta}$ is the new stress diffusion term defined for each stress component as

$$D_i^{\alpha\beta} = 2\zeta h c_0 \sum_j \psi_{ij}^{\alpha\beta} \frac{\mathbf{x}_{ij} \cdot \nabla_i W_{ij}}{|\mathbf{x}_{ij}|^2 + 0.01h^2} \frac{m_j}{\rho_j} \tag{26}$$

Eq. (26) is evaluated in particle interactions and contributes to the stress integration.

There are several options to construct the diffusion operator. The most easily applied form is the one proposed by Molteni and Colagrossi (2009) which adopts the SPH Laplacian operator by Morris et al. (1997). For the diffusion of stress field, the formula by Molteni and Colagrossi (2009) can be modified as,

$$\psi_{ij}^{\alpha\beta} = \sigma_i^{\alpha\beta} - \sigma_j^{\alpha\beta} \tag{27}$$

As shown by Antuono et al. (2012), the use of Eq. (27) leads to,

$$D_i^{\alpha\beta} \;=\; 2\nabla\sigma_i^{\alpha\beta}\nabla\Gamma_i + \Gamma_i\nabla^2\sigma_i^{\alpha\beta} + \mathrm{O}(h) \tag{28}$$

where

$$\Gamma_i = \sum_{j=1}^{N} W_{ij}V_j, \nabla\Gamma_i = \sum_{j=1}^{N} \nabla_i W_{ij}V_j \tag{29}$$

It can be seen that Eq. (28) involves a Laplacian operator that acts as the diffusion term, and also a first-order differential operator. For the particle inside the bulk domain $\nabla\Gamma_i \approx 0$, the first term vanishes and the diffusion term can well approximate the Laplacian of the stress field. However, it presents a problem for the particle with incomplete kernel support such as those close to the free surface, in which the first term is non-zero. Consequently, the formulation by Molteni and Colagrossi (2009) is no longer consistent near the free surface, causing a change of stress along the free surface. The resulting effect is not obvious for a dynamic condition since the dynamic stress involved could be many orders of magnitude higher than this term, but it is important for static cases as will be shown later.

To recover a global convergence, the diffusion operator expressed in Eq. (27) can be corrected according to Antuono et al. (2010), which reads,

$$\psi_{ij}^{\alpha\beta} = \left(\sigma_i^{\alpha\beta} - \sigma_j^{\alpha\beta}\right) - \frac{1}{2}\left(\left\langle\nabla\sigma^{\alpha\beta}\right\rangle_j^L + \left\langle\nabla\sigma^{\alpha\beta}\right\rangle_i^L\right)\cdot\mathbf{x}_{ij} \tag{30}$$

where $\left\langle\nabla\sigma^{\alpha\beta}\right\rangle_j^Land\left\langle\nabla\sigma^{\alpha\beta}\right\rangle_i^L$ are the renormalised stress gradients defined as

$$\left\langle\nabla\sigma^{\alpha\beta}\right\rangle_j^L = \sum_j \left(\sigma_j^{\alpha\beta} - \sigma_i^{\alpha\beta}\right)\boldsymbol{L}_i\nabla_i W_{ij}\frac{m_j}{\rho_j} \tag{31a}$$

$$\boldsymbol{L}_i = \left(\sum_j \left(\mathbf{x}_j - \mathbf{x}_i\right) \otimes \nabla_i W_{ij}\frac{m_j}{\rho_j}\right)^{-1} \tag{31b}$$

It can be seen that with the use of Eq. (30), the first-order differential term vanishes, and the resulting diffusion term can reproduce the Laplacian of stress within the whole domain. Nevertheless, there is a relatively large computational cost for the evaluation of renormalised stress gradients.

In this study, following the idea of the diffusion algorithm recently proposed by Fourtakas et al. (2019), Eq. (27) is modified to restore consistency within the whole domain while avoiding computing the

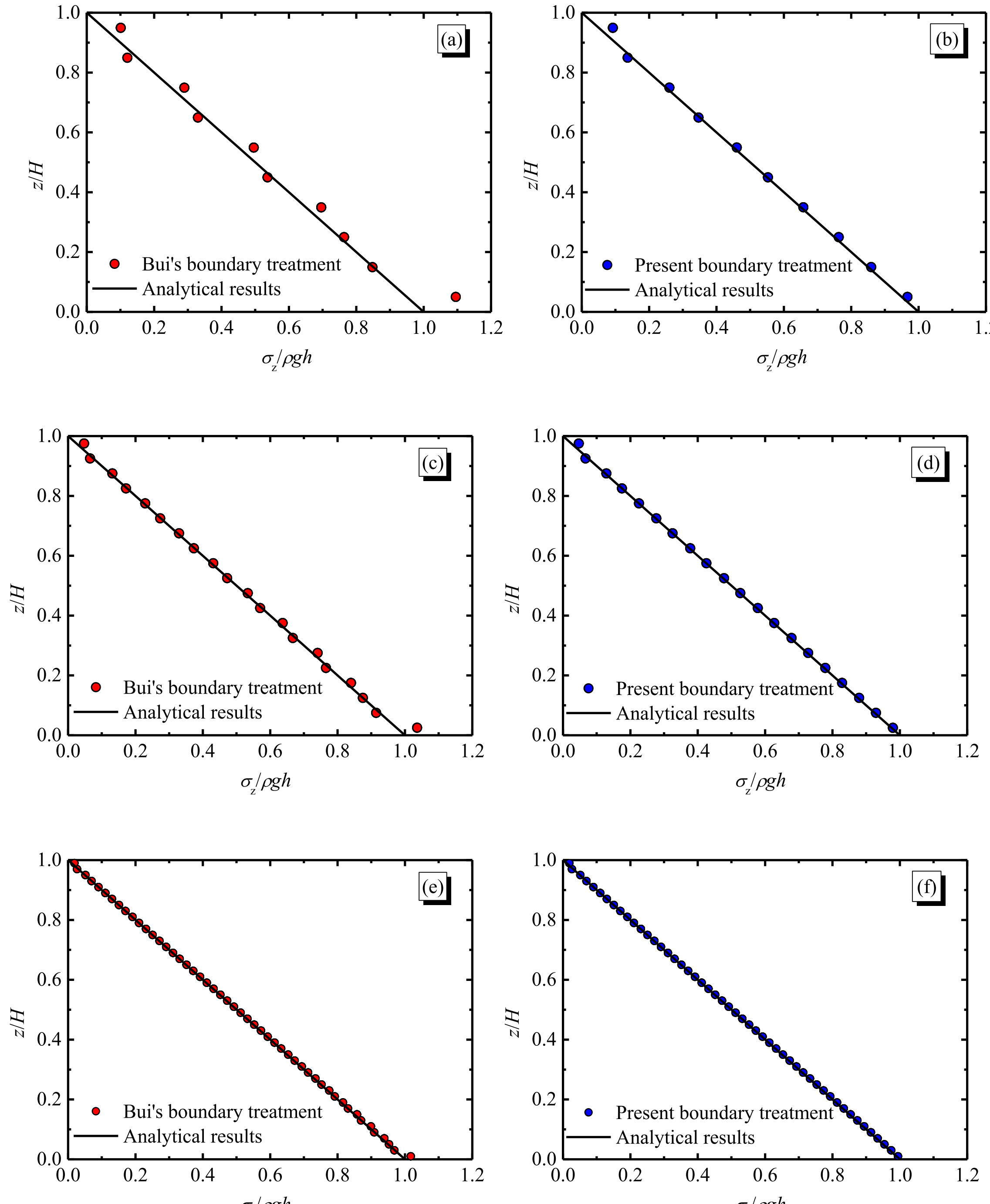

Fig. 8. Normalised vertical stress profile at final instant ($t = 5$ s) for the different resolutions $dp = 0.1$ (first row), $dp = 0.05$ (second row), and $dp = 0.02$ (third row) using the boundary treatment by Bui et al. (2008) (left) and present boundary treatment (right).

normalised stress gradient. This is achieved by using the dynamic stress rather than the total one in the diffusion term, which is given by,

$$\psi_{ij}^{\alpha\beta} = \sigma_i^{\alpha\beta,D} - \sigma_j^{\alpha\beta,D} \tag{32}$$

Since $\sigma^{\alpha\beta,D} = \sigma^{\alpha\beta} - \sigma^{\alpha\beta,S}$, Eq. (32) can be rewritten as,

$$\psi_{ij}^{\alpha\beta} = \sigma_{ij}^{\alpha\beta} - \sigma_{ij}^{\alpha\beta,S} \tag{33}$$

where the superscript D and S refers to dynamic and static components, respectively.

Assuming the shear stress of soil at static condition is zero, this modification only affects the diffusion operator for diagonal component of the stress tensor. The final form of stress diffusion operator can be expressed as,

$$\begin{cases} \psi_{ij}^{\alpha\beta} = \sigma_{ij}^{\alpha\beta} & \alpha \neq \beta \\ \psi_{ij}^{xx} = \sigma_{ij}^{xx} - K_0\rho_0 g z_{ij} \\ \psi_{ij}^{yy} = \sigma_{ij}^{yy} - K_0\rho_0 g z_{ij} \\ \psi_{ij}^{zz} = \sigma_{ij}^{zz} - \rho_0 g z_{ij} \end{cases} \tag{34}$$

where K_0 is Jaky's earth pressure coefficient at rest that will be defined later in Section 5.

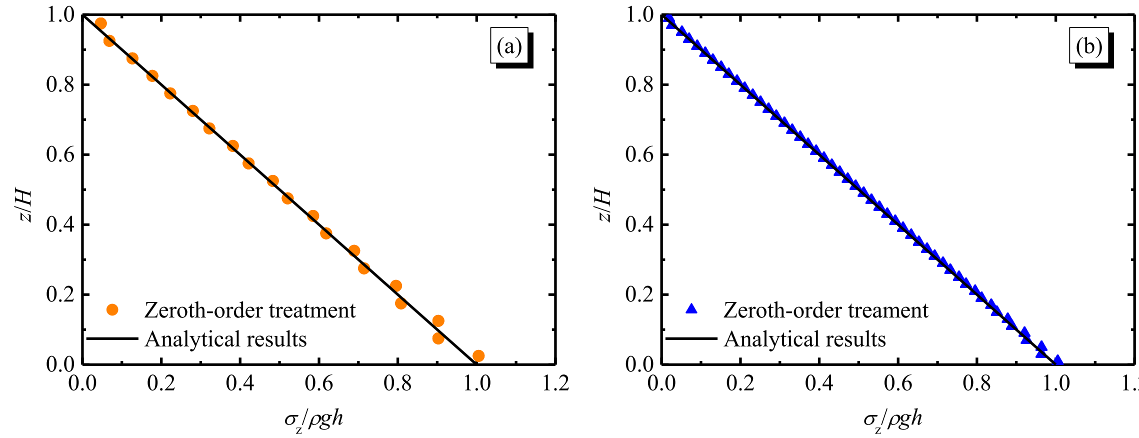

Fig. 9. Normalised vertical stress profile with zeroth-order boundary treatment (a) $dp = 0.05$; (b) $dp = 0.02$.

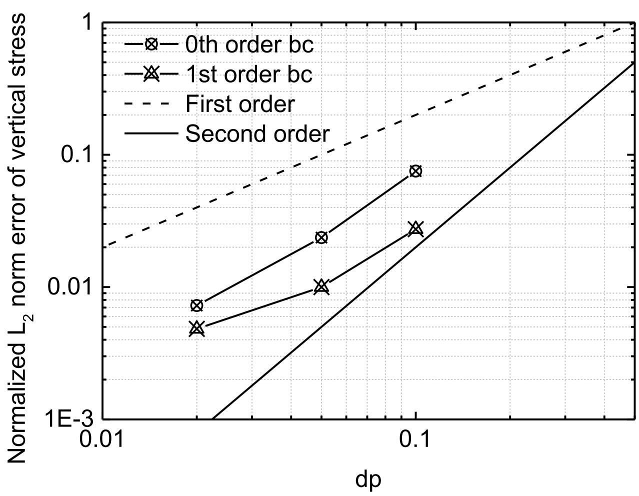

Fig. 10. Static soil column: normalised L_2 error norms of vertical stress at 5 *s*.

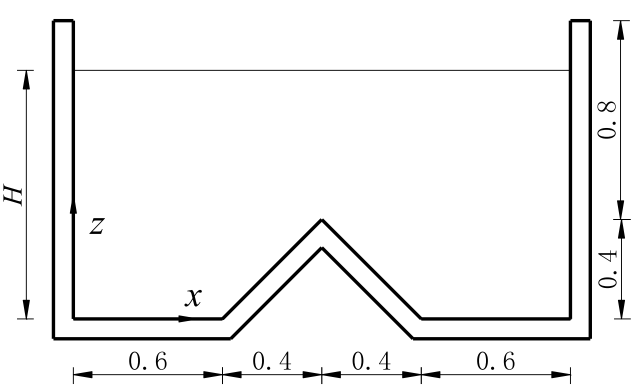

Fig. 11. Schematic diagram for the static soil in a container with wedge.

3.4. Time-stepping scheme

The time integration scheme adopted is an explicit second-order predictor–corrector scheme to integrate Equations (17) (19) (22) (23) and (25) in time. In this scheme, the predictor step evaluates the evolution in time at the middle of the time step by Euler forward stepping, and the obtained values are then corrected using the updated forces in the corrector step. The results obtained from the corrector step are finally used to evaluate the values of these variables at the end of the time step. Further details of the time-stepping scheme can be found in Crespo et al. (2015).

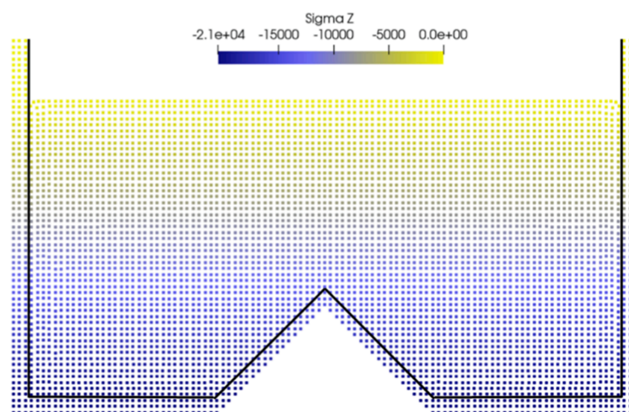

Fig. 12. Static dry soil with wedge: normalized stress contour at 10 s.

The time-stepping schemes is restricted by the CFL (Courant–Friedrichs–Levy) condition, the maximum force term, and the numerical speed of sound. The variable time step is calculated according to (Monaghan and Kos, 1999),

$$\Delta t_f = \min_i\left(\sqrt{h/|\mathbf{f}_i|}\right) \tag{35a}$$

$$\Delta t_{cv} = \min_i \frac{h}{c_s + \max_j\left|\frac{h\mathbf{v}_{ij}\cdot\mathbf{x}_{ij}}{\mathbf{x}_{ij}^2+\eta^2}\right|} \tag{35b}$$

$$\Delta t = C_0 \min\left(\Delta t_{cv}, \Delta t_f\right) \tag{35c}$$

where Δt_f is the time step governed by forces, Δt_{cv} the time step restricted by the Courant and the viscous controls since the artificial viscosity is used in this study, $\mathbf{f}$ is the force per unit mass, c_s is the sound speed of the material, C_0 is the Courant number set to be 0.2.

4. Wall boundary treatment

A common problem of SPH is the kernel truncation of particles near the boundary. Due to the interpolation nature of SPH, the truncated kernel support may lead to inaccurate results and unphysical effects. At present, a particle-based treatment is widely used in which, a set of fictitious particles is created to characterize the boundary of simulated domain. Their properties contribute to the SPH summation of real particles near the boundary to ensure the kernel is fully supported.

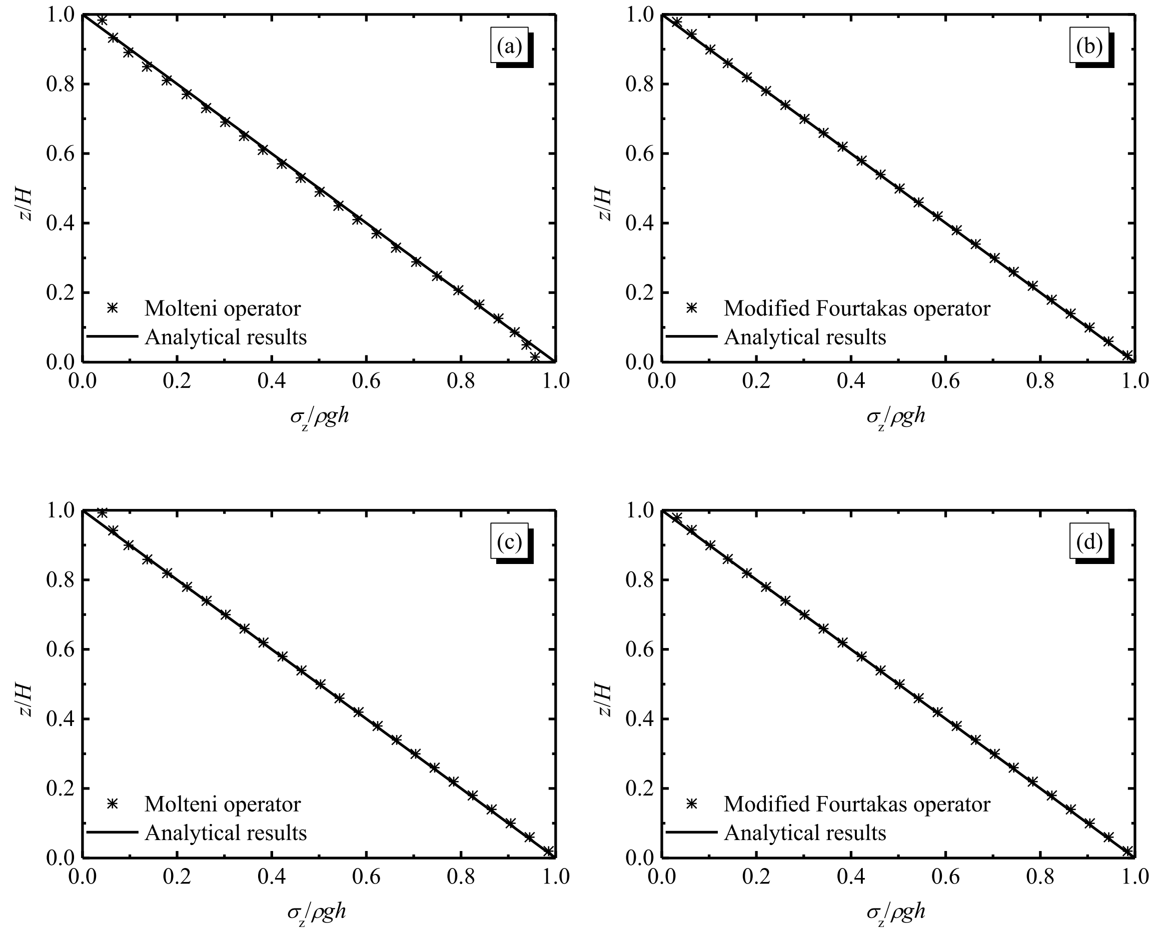

Fig. 13. Static dry soil with wedge: normalised vertical stress profile at 10 s with stress diffusion term ($x = 1.4\ m$). The boundary contribution to the diffusion term is not considered in (a) and (b) whereas it is considered in (c) and (d).

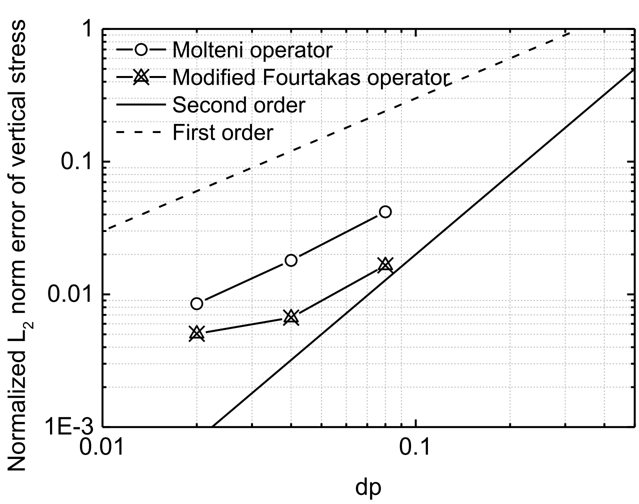

Fig. 14. Static dry soil with wedge: normalised L_2 norm error of vertical stress at 10 s.

In this treatment, the building block of the mDBC (modified dynamic boundary condition) proposed by English et al. (2021), which combines the use of dummy and ghost particles, is introduced with some modifications. The solid wall is discretised with dummy particles. Boundary particles are placed with the same initial particle spacing dp as real particles shown in Fig. 1 but unlike mDBC their properties do not evolve in time.

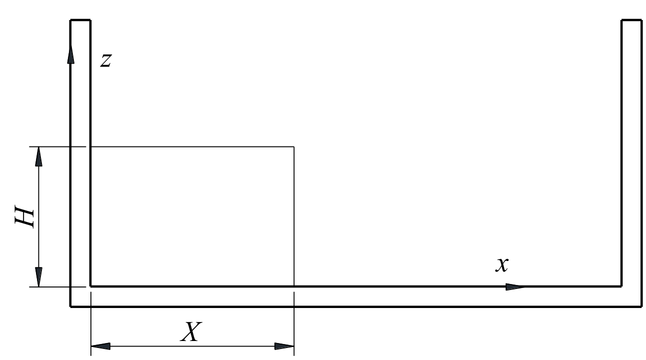

Fig. 15. Schematic diagram of the initial configuration of collapsing granular column problem.

To arrange the boundary particles for arbitrary shaped geometries, two options can be considered. One is to use a cubic lattice to place boundary particles as shown in Fig. 1, and the other is to generate several layers of boundary particles offset from the physical boundary lines. The first option is more suitable for angles of 45° or 90° as a staircase effect can result with other angles. In such a condition, the second option can be adopted.

For each boundary particle, a ghost node (or interpolation point) is projected into the soil domain in a similar manner to Marrone et al. (2011). The position of a ghost particle is calculated according to the direction of the boundary normal and the normal distance of boundary particle to the boundary surface. For a flat or curved surface, this

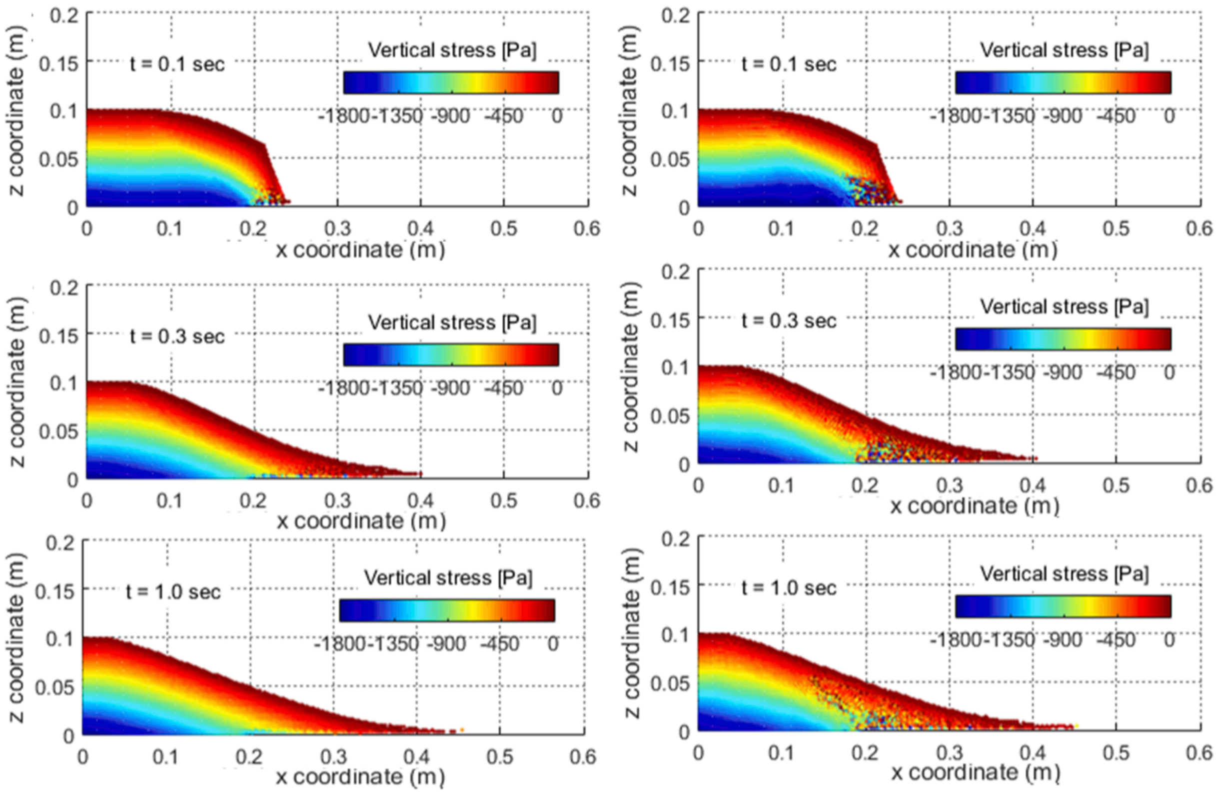

Fig. 16. Vertical stress contour of 2-D granular flow with the diffusion term (left column) and without diffusion term (right column).

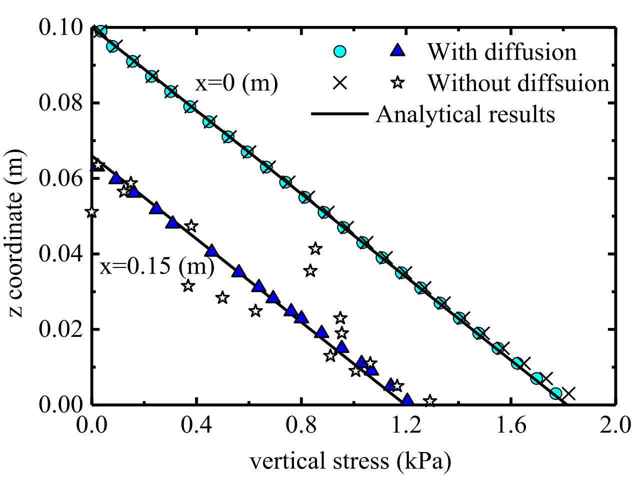

Fig. 17. Vertical stress profile of the final deposit.

technique is straightforward while it presents some difficulties for corners and may require the ghost node to be projected through the corner point according to the distance and direction of the boundary particle to the corner.

The velocity and stress of ghost nodes (denoted by the subscript g) are interpolated from neighbour soil particles, and then feedback to the corresponding boundary particles (denoted by the subscript b). The stress and stress gradients of a ghost particle are computed using the first-order consistent SPH interpolant proposed by Liu and Liu (2006),

$$\mathbf{A}_g\cdot\begin{bmatrix}\sigma_g^{\alpha\beta}\\ \partial_x\sigma_g^{\alpha\beta}\\ \partial_y\sigma_g^{\alpha\beta}\\ \partial_z\sigma_g^{\alpha\beta}\end{bmatrix}=\begin{bmatrix}\sum_j\sigma_j^{\alpha\beta}W_{gj}V_j\\ \sum_j\sigma_j^{\alpha\beta}\partial_xW_{gj}V_j\\ \sum_j\sigma_j^{\alpha\beta}\partial_yW_{gj}V_j\\ \sum_j\sigma_j^{\alpha\beta}\partial_zW_{gj}V_j\end{bmatrix} \tag{36}$$

where the renormalisation matrix $\mathbf{A}_g$ is

$$\mathbf{A}_g=\begin{bmatrix}\sum_j W_{gj}V_j & \sum_j(x_j-x_g)W_{gj}V_j & \sum_j(y_j-y_g)W_{gj}V_j & \sum_j(z_j-z_g)W_{gj}V_j\\ \sum_j\partial_xW_{gj}V_j & \sum_j(x_j-x_g)\partial_xW_{gj}V_j & \sum_j(y_j-y_g)\partial_xW_{gj}V_j & \sum_j(z_j-z_g)\partial_xW_{gj}V_j\\ \sum_j\partial_yW_{gj}V_j & \sum_j(x_j-x_g)\partial_yW_{gj}V_j & \sum_j(y_j-y_g)\partial_yW_{gj}V_j & \sum_j(z_j-z_g)\partial_yW_{gj}V_j\\ \sum_j\partial_zW_{gj}V_j & \sum_j(x_j-x_g)\partial_zW_{gj}V_j & \sum_j(y_j-y_g)\partial_zW_{gj}V_j & \sum_j(z_j-z_g)\partial_zW_{gj}V_j\end{bmatrix} \tag{37}$$

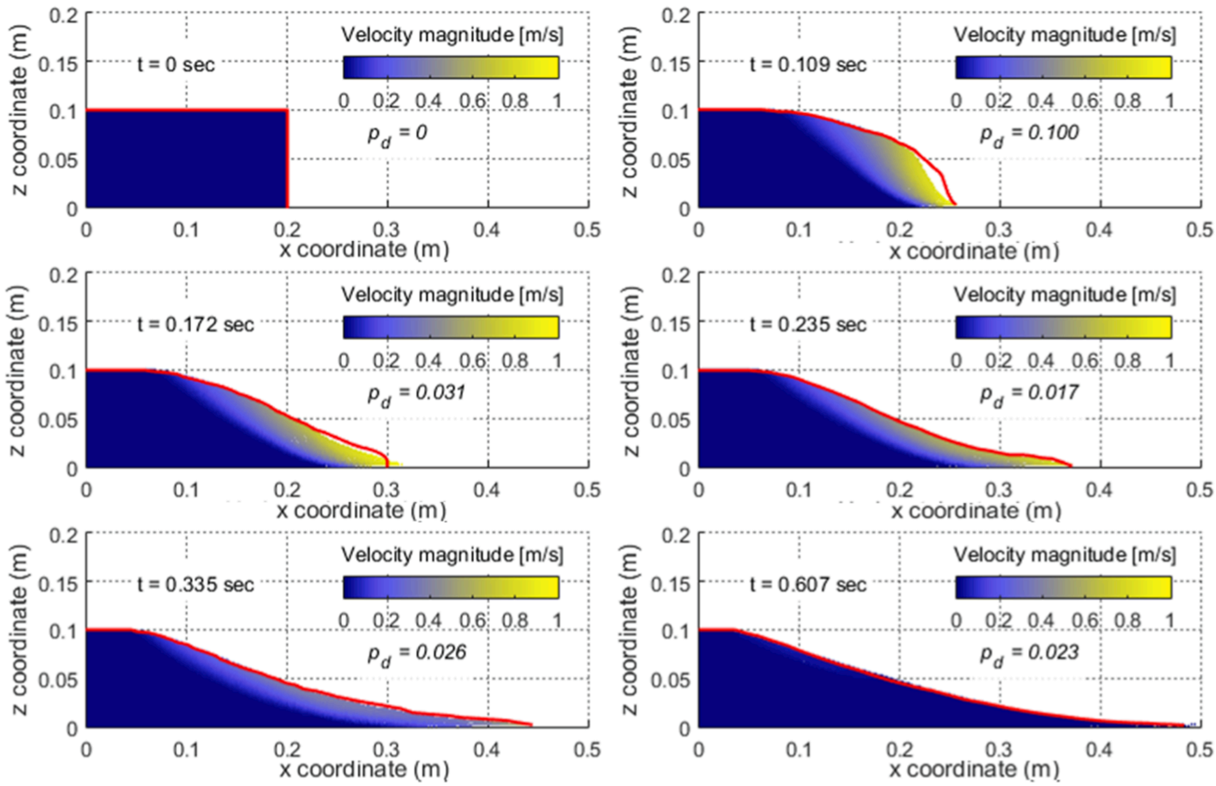

Fig. 18. Comparison between the numerical simulation and the experiment for 2-D granular column collapse progress (red line denotes the experimental profile, a = 0.5). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

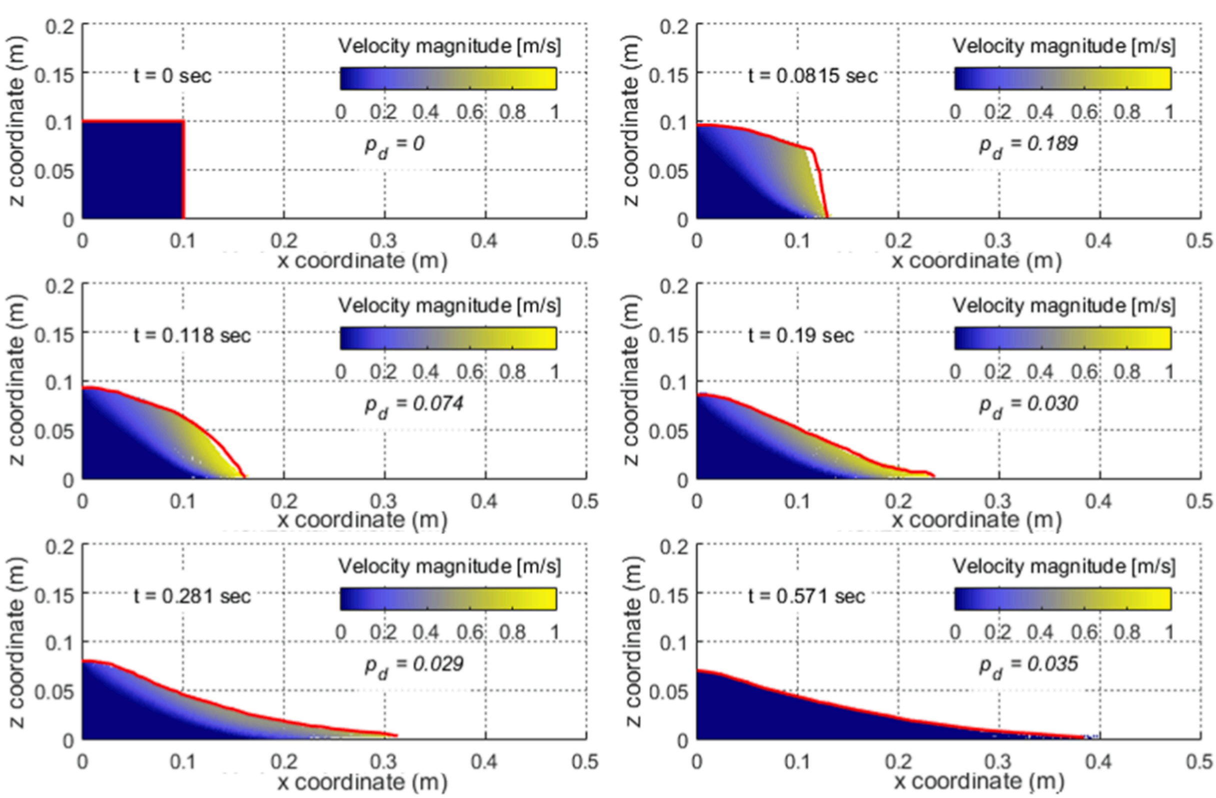

Fig. 19. Comparison between the numerical simulation and the experiment for 2-D granular column collapse (red line denotes the experimental profile, a = 1.0). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

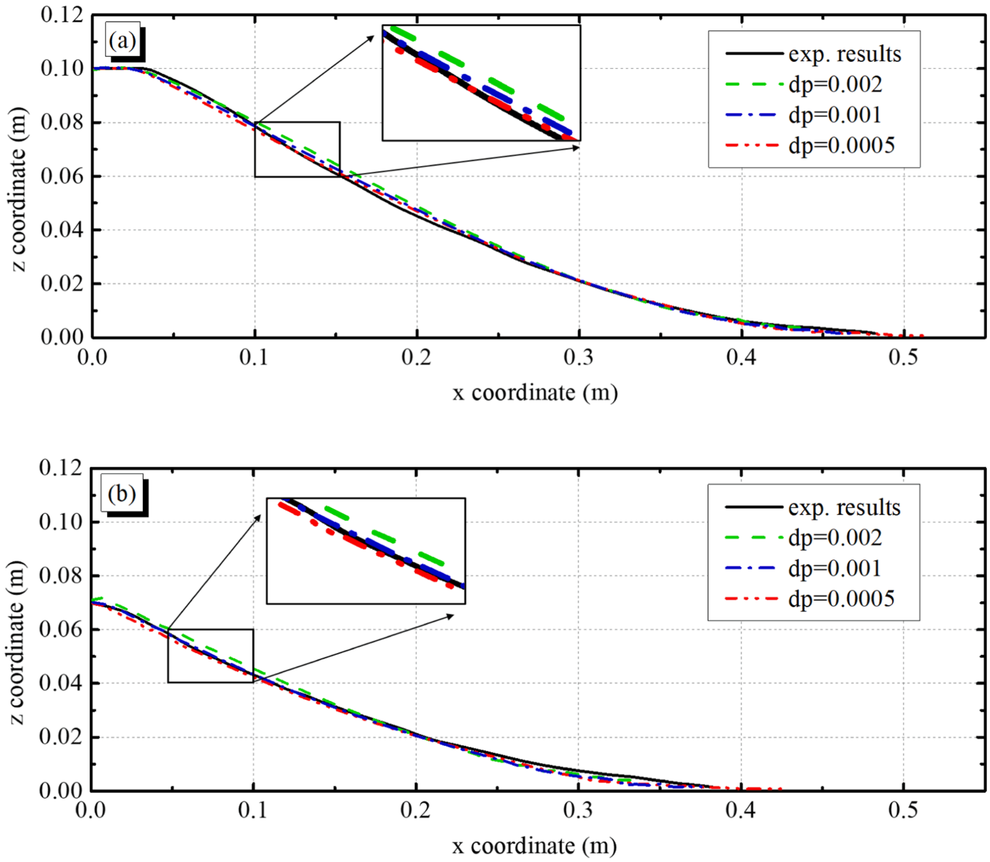

Fig. 20. Comparisons of the free-surface line between SPH simulation and experimental data (a) $a = 0.5$ and (b) $a = 1.0$.

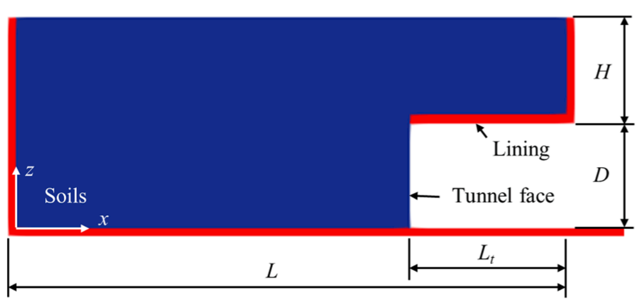

Fig. 21. Computational set-up of the tunnel model (soil particles and boundary particles are marked as blue and red). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
Geometric parameters of the tunnel model.

Parameters	Notations	Values
Total length	L	42 cm
Tunnel length	L_t	12 cm
Tunnel diameter	D	8 cm
Overburden depth	H	8, 16 cm

The final stress value of the boundary particle $\boldsymbol{\sigma}_b$ is evaluated according to the stress and stress gradients at ghost nodes using a Taylor series expansion,

$$\sigma_b^{\alpha\beta} = \sigma_g^{\alpha\beta} + (x_b - x_g)\partial_x\sigma_g^{\alpha\beta} + (y_b - y_g)\partial_y\sigma_g^{\alpha\beta} + (z_b - z_g)\partial_z\sigma_g^{\alpha\beta} \tag{38}$$

It should be noted that although the matrix $\mathbf{A}_g$ is diagonally-dominated and non-singular even for very disordered particle distributions. Yet, it is still possible to be ill-conditioned and the renormalisation matrix collapses to the zeroth order consistent SPH interpolant (Shepard function), given by,

$$\sigma_b^{\alpha\beta} = \sigma_g^{\alpha\beta} = \frac{\sum_j \sigma_j^{\alpha\beta} W_{gj} V_j}{\sum_j W_{gj} V_j} \tag{39}$$

In order to recover the static equilibrium condition near the boundary and produce considerable repulsive forces to prevent particle penetration for this zeroth-order interpolation, the diagonal component of final stress tensor is corrected in a manner similar to Adami et al. (2012) according to the force balance at equilibrium state, which is given by,

$$\sigma_b^{zz} = \sigma_g^{zz} - \rho_0 g z_{bg} \tag{40a}$$

$$\sigma_b^{xx} = \sigma_g^{xx} - K_0 \rho_0 g z_{bg} \tag{40b}$$

$$\sigma_b^{yy} = \sigma_g^{yy} - K_0 \rho_0 g z_{bg} \tag{40c}$$

The above procedure requires the calculation of the inversion of the matrix $\mathbf{A}_g$ and the stress summation vector, in which a single extra loop is needed. The computational overhead is similar with the boundary treatment using renormalization technique (e.g, Peng et al. 2019, Yang et al. 2020) but it is slightly larger than those using the uniform stress technique (e.g., Bui et al. 2008). However, it can create smoother and more physical stress profile as will be demonstrated at later sections.

To create a free-slip/no-slip boundary condition, the velocity at the ghost nodes is calculated using Shepard corrected summation,

$$v_g^{\alpha} = \frac{\sum_j v_j^{\alpha} W_{gj} V_j}{\sum_j W_{gj} V_j} \tag{41}$$

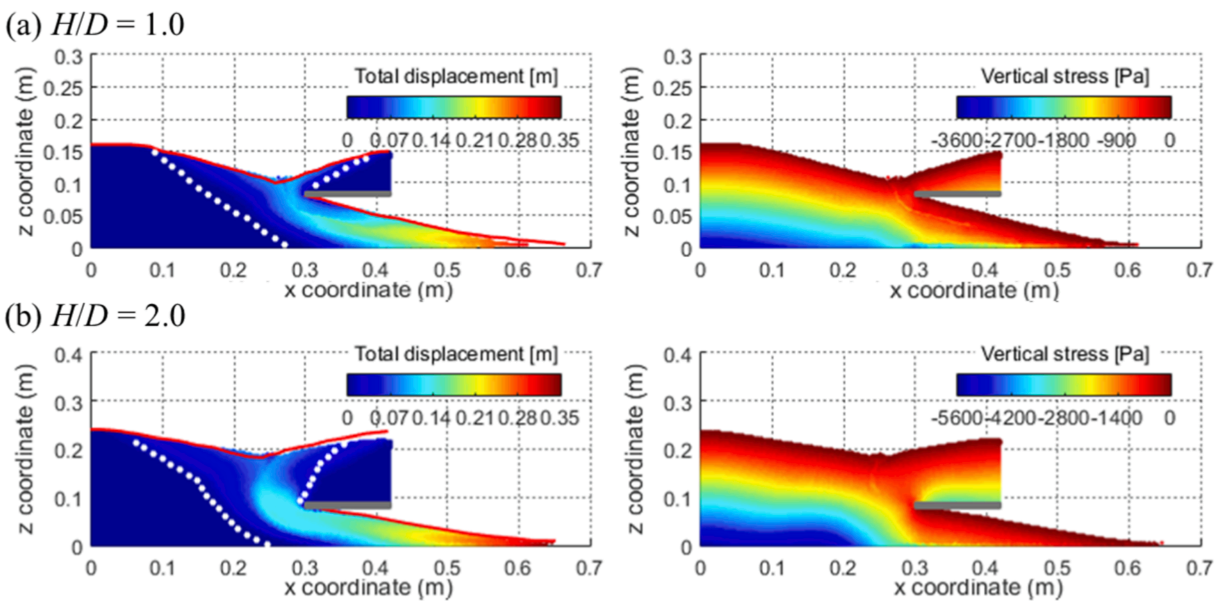

Fig. 22. Comparison between the numerical simulation and the experiment for 2-D tunnel face collapse (a) H/D = 1.0 and (2) H/D = 2.0.

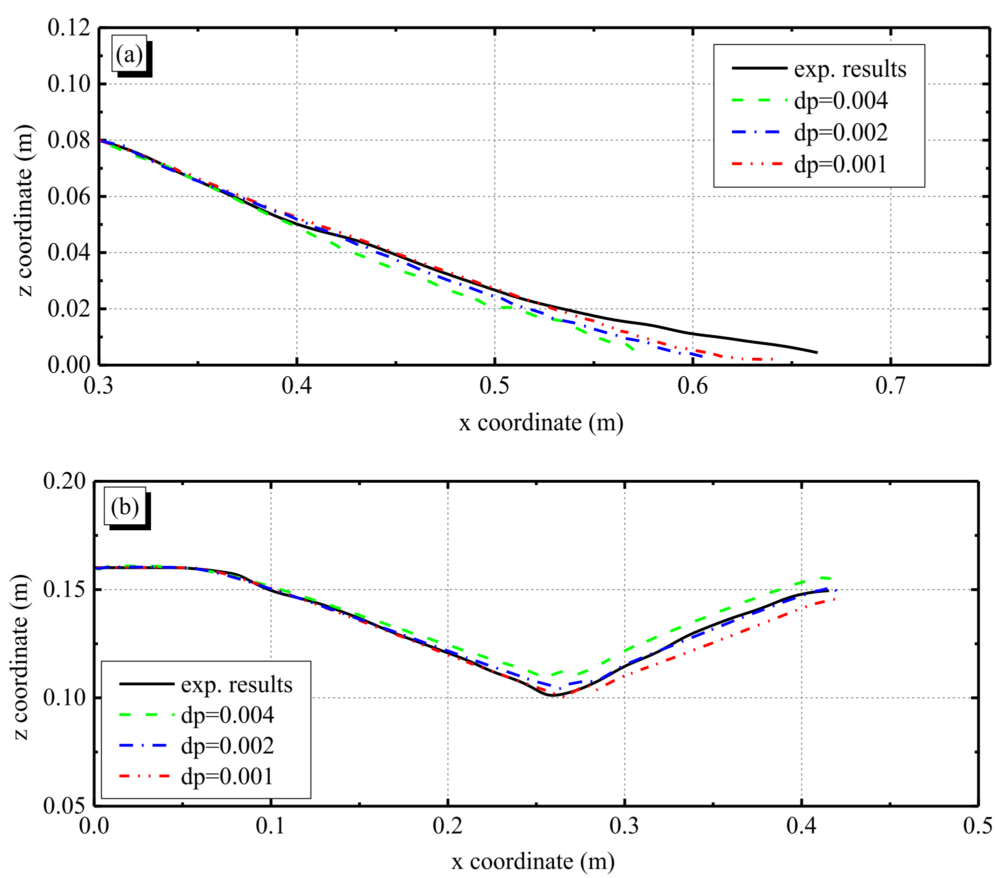

Fig. 23. Comparison with experimental data (H/D = 1.0) for (a) face extrusion and (b) ground surface.

To guarantee a no-slip condition at the boundary interface, the velocity of ghost nodes is assigned to corresponding boundary particles with reversed direction,

$$v_b^\alpha = -v_g^\alpha \tag{42}$$

For a free-slip condition, the boundary particle obtains the ghost node tangential velocity whereas the normal ghost node velocity is reversed,

$$v_b^\alpha = v_g^\alpha - 2v_{g.n}^\alpha \tag{43}$$

where the subscript n denotes the normal component.

Similar to Bui et al. (2008), the density of boundary particles is set to remain constant at the reference density ρ_0.

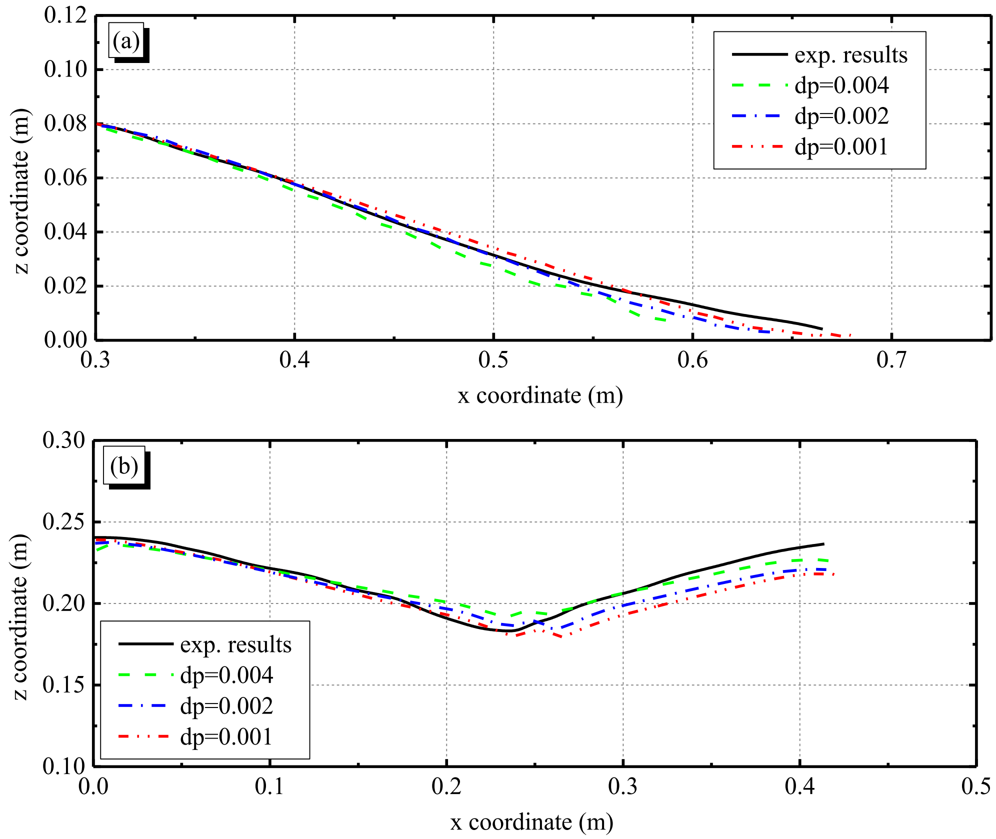

Fig. 24. Comparison with experimental data (H/D = 2.0) for (a) face extrusion and (b) ground surface.

It's worth noting that the proposed boundary condition is different from a mirror boundary condition that projects domain particles into boundary, in which the normal vectors or position of mirror boundary particles need to be re-evaluated every time step. As we mirror boundary particles to the domain and the boundary particles are fixed in space in present methodology, there is no additional efforts for computing the position of mirrored ghost nodes during the simulation (position of ghost nodes are fixed). The computations of the position of ghost nodes need to carefully evaluate the normal vectors at the beginning of the simulation, which slightly increases the efforts for the initialisation of simulation, but the resulting boundary treatment gives improved accuracy and consistency.

In addition, the periodic boundary condition (PBC) and the dynamic boundary condition (DBC) are also used in this study. PBC is adopted for the modelling of Couette flow and static dry soil column to produce a uniform deformation. The DBC is used for the simulation of simple shear test solely to create a pure shear deformation. When using the DBC, dummy particles are employed for the discretization of wall boundary. When adapted for our scheme, density, stress and strain of these boundary particle share the same governing equations with soil particles and evolve with time, but the position of boundary particles is either fixed in space or moving with the prescribed wall boundary velocity.

5. Validation of the proposed numerical model

The numerical scheme described above is implemented in the CPU version of the open-source SPH solver DualSPHysics (Crespo et al 2015) and validated against several test cases, including Couette flow, simple shear test, and static dry soil column. Finally, the proposed method is applied to study 2-D granular column collapse and tunnel face collapse. In this study, the Jaky's equation for the earth pressure coefficient at rest, i.e., $K_0 = 1\text{-}\sin\varphi$, is adopted for the numerical diffusive term expressed in Eq. (32) and zero-th order boundary treatment.

Note that a relatively large h/dp is used in all cases to improve the accuracy of the results. In addition, this selection is beneficial for the study of convergence behaviour since total error is operated in the smoothing-error dominated regime (Quinlan et al. 2006), and it is possible to achieve convergence with the refinement of dp.

5.1. Element shear test

5.1.1. Implementation of no-slip condition: The Couette flow

The Couette flows of linear elastic material are simulated to examine the ability of the new boundary treatment to produce a no-slip boundary condition. Fig. 2 shows the numerical set-up, which consists of two horizontal boundary walls placed at z = -0.5 m and z = 0.5 m, respectively.

The bottom wall is fixed whereas the top one moves with a velocity $v_x = 0.001$ m/s. The initial particle spacing is set to 0.04, 0.02, and 0.01 m, resulting in 25, 50, and 100 particles across the height of the domain. Periodicity is applied horizontally to ensure a uniform shear deformation. The parameters adopted in the simulation are listed in Table 1. A relatively large artificial viscosity parameter is used in this case to dissipate the numerical oscillations in the calculation of the momentum and to stabilize the simulation. It is finally noted that gravity is not considered in the present simulation.

Fig. 3 shows the velocity field in the x direction at $t = 5.0$ s for the coarsest resolution. The particle velocities at steady state after $t = 5.0$ s are plotted against the corresponding analytical solution in Fig. 3(a). From the comparison it can be seen that a satisfactory agreement has been achieved with the L_2 norm error less than 0.75%. The convergence behaviour of velocity at steady state is shown in Fig. 4 with the order of convergence of approximately 1.21, indicating that the no-slip condition is adequately reproduced with the new boundary treatment.

5.1.2. Validation of Drucker-Prager constitutive model: Simple shear test

To validate the implementation of the Drucker-Prager elasto-plastic model, the simulation of a series of simple-shear tests are conducted and compared against analytical solutions. Fig. 5 shows a sketch of simple-shear test set-up and its numerical implementation. The numerical domain is square shaped with length of 0.1 m. The soil particles are placed in the centre of the simulated domain with the boundary particles enclosing the central area. Since the relationship between stress and strain is independent of the particle spacing, a relatively coarse particle resolution is adopted. The initial particle distance dp is 0.01 m, resulting in a total of 100 particles. The parameters and soil properties adopted in the simulation are listed in Table 1. The confining pressure is applied by assigning an initial mean principal stress for all the particles. Three levels of initial confining stress of 150, 225, and 300 kPa are tested.

A constant velocity profile is prescribed for the boundary particles and maintained throughout the simulation. DBC is adopted for this application to enable the pure shear deformation of boundary particles. It has been a common choice for testing constitutive model in SPH (e.g., Nonoyama et al. 2015; Zhao et al. 2019). Our aim of this test case is not to investigate the effectiveness of the wall BCs but the validity of the stress–strain relationship. The new BCs are demonstrated in other cases. In the horizontal direction, the velocity of the boundary particles is given by of $v_x = 0.01z$ where z denotes the vertical distance from the central axis. A zero-vertical velocity, $v_z = 0$ m/s, is prescribed for the boundary particles, corresponding to a constant shear rate of $\dot{\gamma}_{xz} = 1.0\%/s$. The gradient renormalization technique, Eqs. (36)-(38), is applied in the calculation of strain rate equation to create an accurate prediction of strain rate. The calculated stress and strain of the single particle at the centre of the specimen are recorded throughout the simulation.

The simulations are conducted assuming zero gravity. The Jaumann rate is not considered in this simulation since the deformation of the specimen is relatively small. The stress path and shear stress–strain relationships at the centre of soil domain are plotted in Fig. 6 together with the analytical solutions. From Fig. 6(a) it can be seen the initial confining stress keeps constant until the stress state reaches the yielding line. The stress state remains on the yield surface after yielding, indicating the consistency condition for elastic–plastic model is correctly reproduced. The obtained stress–strain relationship (Fig. 6b) agrees well with the analytical solution, demonstrating that the constitutive model implemented in DualSPHysics is able to produce the appropriate stress state with high accuracy.

5.2. Static dry soil

5.2.1. Implementation of boundary conditions: The soil column test

The simulation of the static soil column with level ground is performed to study the performance of the presented boundary condition under stress increment. The model setup is shown in Fig. 7(a). Periodic boundaries are applied to both sides to reproduce the infinite lateral extent of the soil column. The depth of the soil layer is $H = 1$ m. Three sets of particle resolutions of 0.1, 0.05 and 0.02 m are adopted, resulting in 10, 20 and 50 particles along the column height, respectively. The no-slip solid boundary condition proposed by Bui et al. (2008) is also used for comparison with the proposed boundary condition treatment. Simulations are conducted for 5 s of physical time. More details of the adopted model and parameters can be found in Table 1.

As observed in previous studies (e.g., Bui and Fukagawa, 2013; Mao et al. 2017), the sudden application of a force (e.g., gravitational loading) to the soil body results in high-frequency stress oscillation. To settle down the system to an equilibrium state, the damping term (equation 52) developed by Bui and Fukagawa (2013), with damping coefficient of $\xi = 0.02$, is introduced into the momentum equation as,

$$\left\langle \frac{\mathrm{d}v^{\alpha}}{\mathrm{d}t} \right\rangle_i = \sum_{j=1}^{N} m_j \left(\frac{\sigma_i^{\alpha\beta} + \sigma_j^{\alpha\beta}}{\rho_i \rho_j} + \Pi_{ij}\delta^{\alpha\beta} \right) \frac{\partial W_{ij}}{\partial x_i^{\beta}} + g_i^{\alpha} + d_i^{\alpha} \tag{44}$$

where d is the damping force term given by,

$$d_i^{\alpha} = -\xi \sqrt{\frac{E}{\rho h^2}} v_i^{\alpha} \tag{45}$$

Note that, the damping process is only considered when obtaining the stress profile at static condition. For a dynamic case with particle motion, the damping term should be switched off as it will lead to unphysical energy dissipation.

Particles coloured by the vertical stress in the soil column at the final instant are shown in Fig. 7(b). Fig. 8 shows the final distribution of vertical stress with depth for different boundary treatments and particle resolutions.

It can be seen that for the treatment by Bui et al. (2008), the stress distribution near the boundary diverges from the analytical solution. This discrepancy is clear for a large particle spacing. By contrast, the proposed boundary treatment results in a stress distribution closer to its analytical value even for the lowest resolution of 0.1 m.

Furthermore, the zero-th order option of the presented boundary condition using Eq. (39) is also tested. The comparison between numerical and analytical results is shown in Fig. 9. Similar to the boundary treatment by Bui et al. (2008), there are fluctuations in the stress field near the boundary when using the presented zero-th order boundary treatment since the stress value at the boundary is not accurately solved. In spite of that, the proposed treatment can still provide sufficient accuracy when the first-order interpolation cannot be attained due to ill conditioned matrices of Eq. (37).

In addition, the normalized L_2 error norms of vertical stress for three different particle resolutions are presented in Fig. 10. Clearly, both the zeroth-order and first-order boundary treatment show a convergent behaviour with the refinement of particle spacing.

5.2.2. Validation of complex geometries: The container with a wedge

The simulation of static soil in a container with wedge is performed to demonstrate the ability of the presented boundary condition to handle the complex geometry, such as the discontinuous points and slopes within the domain. As shown in Fig. 11, the container has dimension of 2×1.2 m and a triangular wedge is placed at the centre with a height of 0.4 m and $\pi/4$ rad angle. The initial soil height is set to be $H = 1$ m with an initial soil density of 2100 kg/m^3. The initial particle spacings for this case are taken as $dp = 0.08$, 0.04 and 0.02 m. The free-slip condition is applied for the side wall, and no-slip condition is enforced for the bottom wall using the proposed boundary conditions. Table 1 lists the parameters adopted. To obtain an analytical solution with such a complex boundary geometry, the elastic constitutive model is used in this case and the shear modulus is set as zero (hence the deviatoric stresses are not considered), making the materials behave like a fluid. The proposed stress diffusion term is also applied for this application, together with the use of diffusion operator proposed by Molteni and Colagrossi (2009) for comparison. Further, the damping algorithm of Eq. (44) is used to settle down initial numerical oscillations.

Fig. 12 shows the normalized vertical stress field at the final instant. Evidently, the use of present boundary treatment produces a smooth and accurate stress profile, not only in the flat surface but also in the corners.

The vertical stress profile of soil particles at $x = 1.4\,m$ using the stress diffusion term with Molteni and Colagrossi operator and the formulation proposed in this study is shown in Fig. 13. In Fig. 13(a), the diffusion term is only applied for the soil-soil interaction. The artificial diffusion term of Molteni and Colagrossi operator shows a variation of stress near both the free-surface and wall boundary, which is caused by the lack of consistency of the operator with truncated kernel support at the surface. If the contribution of boundary particles is considered to complete the kernel support of the diffusion term near the boundary, the variations of stress can be improved as shown in Fig. 13(c). However, it should be noted that the correct diffusion near the boundary requires the boundary condition to be of sufficient accuracy. For the boundary condition where the stress value is not accurately resolved, it is possible to induce unphysical diffusion along the soil-boundary interface if the soil-boundary interaction is included in the diffusion term.

This problem can be improved by using the proposed modified diffusion operator of Eq. (34). In Fig. 13(b) and (d), good agreement between numerical and analytical results are achieved regardless if the boundary contribution considered. Evidently, the use of the proposed diffusive algorithm is able to restore the consistency for particles with incomplete kernel support.

A convergence study on the normalised L_2 error of vertical stress is shown in Fig. 14. Although the orders of convergence of using these two diffusion operators are very close, the reduction of error by using the proposed modified Fourtakas operator is evident.

5.3. Analysis of large deformation problem

5.3.1. 2-D granular column collapse

This section presents results from a 2-D granular column collapse used to examine the ability of the proposed scheme to simulate large deformation problems in the presence of a free-surface. Fig. 15 shows the initial geometry and boundary condition of the test case. The granular column has the dimension of 20 cm in length and 10 cm in height. The vertical wall is assumed to be free-slip, while the bottom wall is treated as a no-slip boundary.

The initial particle spacing is set to 0.002 m, resulting in 5000 soil particles. The parameters and constitutive properties adopted in the simulation are listed in Table 1.

Fig. 16 shows the evolution of stress profile of the granular flows with and without diffusion term at different time instants. In both cases, artificial viscosity is adopted. It can be observed that without the use of diffusion term, the short-length-scale noise is developed especially in the region which undergoes large deformations. These fluctuating stress distributions can be significantly improved by introducing numerical diffusion in stress, resulting in a noise-free smooth stress distribution. In addition, the surface profiles of two configurations at each instant are nearly identical, indicating the use of present diffusion term does not alter the physics of the flow.

The predicted vertical stresses at the locations of $x = 0$ m and $x = 0.15$ m are plotted in Fig. 17 against the theoretical solution. It can be seen that the simulation without the diffusion term provides smooth and accurate stress profile within the undisturbed region, but the stress distribution is scattered in the highly deformed region. This can lead to discrepancies for cases that the stress/force prediction at the location experiencing large deformation, such as for transient cases i.e., evaluation of the impact of debris. The spurious stress profile is improved with the present SPH scheme, in which a smooth and accurate prediction of stress can be achieved for both undisturbed and deformed regions, thus allowing better insight into the physics of the failure processes. The slight difference can be attributed to the change of bulk density during the flow while the reference solution still uses the initial bulk density.

It is worth noting that a smooth stress profile over the entire domain can also be attained with the stress regularisation technique recently proposed by Nguyen et al. (2017) as demonstrated in their work. This involves applying an MLS-corrected SPH interpolation of stress every 5 time steps, in which an extra particle sweep is needed. By contrast, the proposed diffusion term is easy to implement and does not require any stress interpolation procedures, which has benefits in terms of simplicity and computational efficiency. In addition, since the diffusion term contributes to the stress equation, which is time-integrated, for a problem with relatively small time-step or a long-time simulation, there would not be a problem of over-filtering as may be encountered for the stress regularisation technique.

To evaluate the accuracy of numerical scheme, a series of numerical tests is performed and compared with the experimental results presented by Nguyen et al. (2017). The adopted material properties are from the original work and are summarised in Table 1. Two sets of initial aspect ratio a (the ratio of the initial height to the initial width) of 0.5 and 1.0 are investigated, corresponding to the size of 20×10 cm and 10×10 cm domain. The boundaries are assumed to be non-slip at the bottom and the side wall. The particle resolutions for this test series are $dp = 0.002$, 0.001 and 0.0005 m, resulting in 5000, 20000, and 80,000 soil particles.

Figs. 18 and 19 show the velocity profiles of the granular flows with the discretised medium obtained from the simulation at several time instants. The free-surface line from the experiment is also plotted for comparison. The difference between numerical and experimental results is quantified as Gomez-Gesteira et al. (2010),

$$P_{\mathrm{d}} = \left(\sum_{j}^{N} \left(f_j^{\mathrm{num}} - f_j^{exp} \right)^2 \Big/ \sum_{j}^{N} \left(f_j^{exp} \right)^2 \right)^{1/2} \tag{46}$$

where P_{d} is the phase difference between both signals, in which perfect agreement leads to $P_{\mathrm{d}} \rightarrow 0$, f is the variable under consideration (position of the free-surface particles in this case) and the subscripts denote experimental or numerical values.

It can be seen that the movement starts at the granular front after the flow is released, and it develops progressively towards the free surface when the toe front stops. There exists a shear band interface where the particles below remain stationary while those above move outwards. The evolutions of the free-surface profile are well predicted, with the phase difference P_{d} generally on the order of 1%. In addition, it can be found that for the larger aspect ratio $a = 1.0$, the column collapses completely, leading to the final height being lower than its initial value, while for case with $a = 0.5$, the collapse follows a shear failure of the column edge, leaving an undisturbed region found at the inner part with the final height equal to the initial one. These results indicate that the failure mechanism of these two sets of aspect ratios is well captured by the proposed model.

Finally, a comparison of the final arrangement of the deposit against the experimental results for all the three particle resolutions is presented in Fig. 20. It can be seen that the numerical results agree well with experimental results for both $a = 0.5$ and 1.0, and, as the resolution of the particle spacing increases, the predicted free-surface line shows to convergence to the experimental results.

In addition, it can be found that the toe front position of the final deposit increases with increasing particle resolution, because the constitutive equations are continuous and particle-size independent. Once the material properties are specified, as long as the particle spacing decreases, the height of the toe front keeps decreasing while the extent of run-out increases, giving a higher resolution for the flows of granular materials with arbitrary grain size. As commonly defined in literature (e.g., Utili et al. 2015, Nguyen et al. 2020), only the collective behaviour of granular flow is considered in the determination of the flow morphology. The final height of the toe front can be assumed to be consistent with the mean grain size of the granular bulk. Thus, for the prediction of final run-out distance from the proposed SPH model, one can use the actual mean grain size in practice to determine the final height of the toe front and also the run-out distance (for example in this case the mean grain size is about 0.002 m, the final run-out distance can be estimated as the x coordinate of the point with the height of 0.002 m).

5.3.2. 2-D tunnel face collapse

In this case, the present numerical scheme is applied to study the deformation and post-failure behaviour of tunnel faces. The geometries and material properties of the tunnel model are defined to be the same as the experiments conducted by Matsuo et al. (2016) for comparison. Two overburden patterns are investigated, with the ratio of overburden depth to the tunnel diameter H/D equals to 1.0 and 2.0. The definition sketch of the numerical model and the applied geometric parameters are shown in Fig. 21 and listed in Table 2, respectively. The lining structure constructed at the tunnel roof is considered as rigid, which is implemented by using several layers of boundary particles. Open-face condition (e.g., the new Austrian tunnelling method) is assumed in this application and thus no face pressure is applied. The boundaries are assumed to be non-slip for the base and horizontal wall (lining structure), while free-slip conditions are enforced at the vertical walls. Table 1 presents the adopted material properties and numerical parameters. The initial particle spacings for this case are set as $dp = 0.004$, 0.002, and 0.001 m.

Fig. 22 shows the comparisons of the final configuration of tunnel model between the experiment and simulation for $H/D = 1.0$ and 2.0 for $dp = 0.002$. Particles are coloured by the total displacement and vertical stress. The red lines denote the measured surface profile. The white dots refer to the measured failure lines that sperate the deformed region from the undisturbed region. The free-surface profile and the failure line are well-predicted by the proposed SPH scheme, except for a slight overestimation of the ground surface subsidence for a large overburden ratio $H/D = 2.0$.

Additionally, it is found that the predicted ground subsidence follows a Gaussian distribution, which is consistent with previous studies (e.g., Marshall et al. 2012). The maximum ground settlement develops at around half of the tunnel diameter ahead of the tunnel faces, and the magnitude of subsidence for the case of $H/D = 1.0$ is lower than that of $H/D = 2.0$, indicating a larger overburn depth, may lead to a shallow settlement trough.

Finally, a comparison of the face extrusion and ground surface settlement against experiment results with 3 particle spacings for $H/D = 1.0$ and $H/D = 2.0$ is presented in Fig. 23 and Fig. 24. Again, the predicted free-surface profile shows close agreement with the experimental results especially as the particle resolutions increases.

6. Conclusions

This paper presents a new SPH methodology for the large deformation analysis of geomaterials. A new no-slip/free-slip boundary treatment using the fixed ghost particle technique together with first-order consistent interpolation is introduced. It is the first time for such a boundary methodology extended for the application of SPH to solve geotechnical problems. This new boundary formulation is applicable to complex shaped geometries and is demonstrated to provide an accurate prediction of stress and velocity. An improvement on the spurious stress field under large deformation, which is a common issue in the SPH modelling of geomaterials, has been proposed by including a numerical diffusion term into the stress. Several approaches to construct the diffusion term have been discussed, and compared with the new diffusive term algorithm. It is shown to be able to smooth out the numerical noise while maintaining the consistency at the boundary. This treatment is easy to implement and does not require additional costly procedures for stress interpolation. Results from the Couette flow, static dry soil in a complex geometry show a convergent and accurate prediction of the velocity and stress profiles, demonstrating the accuracy, robustness and applicability of the proposed techniques. The new scheme is finally used for the simulations of granular column collapse and tunnel face collapse, showing an excellent agreement between numerical and experimental results and providing smooth stress profiles.

CRediT authorship contribution statement

Ruofeng Feng: Conceptualization, Methodology, Software, Validation, Investigation, Writing – original draft, Writing - review & editing. **Georgios Fourtakas:** Conceptualization, Methodology, Software, Writing - review & editing. **Benedict D. Rogers:** Conceptualization, Methodology, Resources, Writing - review & editing. **Domenico Lombardi:** Conceptualization, Resources, Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Generalized form of the elastic-perfectly plastic model

Herein, the general form of the elastic-perfectly plastic model is derived. Initially, the total strain rate tensor is decomposed into two parts: elastic strain rate tensor and plastic strain rate tensor,

$$\dot{\varepsilon}^{\alpha\beta} = \dot{\varepsilon}_e^{\alpha\beta} + \dot{\varepsilon}_p^{\alpha\beta} \tag{A1}$$

where the subscripts e and p denote elastic and plastic component, respectively.

The elastic component can be calculated according to the generalized Hooke's law as follow,

$$\dot{\varepsilon}_e^{\alpha\beta} = \frac{\dot{s}^{\alpha\beta}}{2G} + \frac{\dot{\sigma}^{\gamma\gamma}}{9K}\delta^{\alpha\beta} \tag{A2}$$

where $\dot{s}^{\alpha\beta}$ is the deviatoric stress rate tensor; G and K are the elastic shear modulus and elastic bulk modulus, respectively; $\delta^{\alpha\beta}$ is the Kronecker's delta.

The plastic component can be derived from the plastic flow rule according to,

$$\dot{\varepsilon}_p^{\alpha\beta} = \dot{\lambda}\frac{\partial g}{\partial \sigma^{\alpha\beta}} \tag{A3}$$

where $\dot{\lambda}$ is the rate of plastic multiplier; g is the plastic potential function.

Substituting Eqs. (A2) and (A3) into Eq. (A1), and rearranging the obtained equation, the generic form of the elastic-perfectly plastic model is given by,

$$\dot{\sigma}^{\alpha\beta} = 2G\dot{e}^{\alpha\beta} + K\dot{\varepsilon}^{\gamma\gamma}\delta^{\alpha\beta} - \dot{\lambda}\left[\left(K - \frac{2G}{3}\right)\frac{\partial g}{\partial \sigma^{mn}}\delta^{mn}\delta^{\alpha\beta} + 2G\frac{\partial g}{\partial \sigma^{\alpha\beta}}\right] \tag{A4}$$

where $\dot{e}^{\alpha\beta}$ is the deviatoric strain rate tensor that can be expressed as,

$$\dot{e}^{\alpha\beta} = \dot{\varepsilon}^{\alpha\beta} - \frac{1}{3}\dot{\varepsilon}^{\gamma\gamma}\delta^{\alpha\beta} \tag{A5}$$

The plastic multiplier can be computed by the consistency condition that is defined as,

$$\mathrm{d}f = \frac{\delta f}{\delta\sigma^{\alpha\beta}}\mathrm{d}\sigma^{\alpha\beta} = 0 \tag{A6}$$

where f is the yield function that defines the onset of plastic deformation.

Combining Eq. (A4) and Eq. (A6), the plastic multiplier can be obtained as,

$$\dot{\lambda} = \frac{\frac{\delta f}{\delta\sigma^{\alpha\beta}}\left[2G\dot{\varepsilon}^{\alpha\beta} + \left(K - \frac{2G}{3}\right)\delta^{\alpha\beta}\dot{\varepsilon}^{\gamma\gamma}\right]}{\left(K - \frac{2G}{3}\right)\left(\frac{\partial f}{\partial\sigma^{mn}}\delta^{mn}\frac{\partial g}{\partial\sigma^{mn}}\delta^{mn}\right) + 2G\frac{\partial f}{\partial\sigma^{mn}}\frac{\partial g}{\partial\sigma^{mn}}} \tag{A7}$$

Considering the following identities,

$$\frac{\partial f}{\partial\sigma^{\alpha\beta}} = \frac{\partial f}{\partial I_1}\delta^{\alpha\beta} + \frac{\partial f}{\partial J_2}s^{\alpha\beta} + \frac{\partial f}{\partial J_3}t^{\alpha\beta} \tag{A8}$$

$$\frac{\partial g}{\partial\sigma^{\alpha\beta}} = \frac{\partial g}{\partial I_1}\delta^{\alpha\beta} + \frac{\partial g}{\partial J_2}s^{\alpha\beta} + \frac{\partial g}{\partial J_3}t^{\alpha\beta} \tag{A9}$$

where I_1 is the first principal stress invariant, J_2 and J_3 are the second and third deviatoric stress invariants, respectively. The term $t^{\alpha\beta}$ is defined as follows,

$$t^{\alpha\beta} = s^{\alpha m}s^{m\beta} - \frac{2}{3}J_2\delta^{\alpha\beta} \tag{A10}$$

the generalized form of stress–strain relationship in terms of stress invariants for an elastic-perfectly plastic material is given by,

$$\dot{\sigma}^{\alpha\beta} = 2G\dot{e}^{\alpha\beta} + K\dot{\varepsilon}^{\gamma\gamma}\delta^{\alpha\beta} - \dot{\lambda}\left[3K\frac{\partial g}{\partial I_1}\delta^{\alpha\beta} + 2G\left(\frac{\partial g}{\partial J_2}s^{\alpha\beta} + \frac{\partial g}{\partial J_3}t^{\alpha\beta}\right)\right] \tag{A11}$$

$$\dot{\lambda} = \frac{1}{H}\left[3K\frac{\partial f}{\partial I_1}\delta^{\alpha\beta} + 2G\left(\frac{\partial f}{\partial J_2}s^{\alpha\beta} + \frac{\partial f}{\partial J_3}t^{\alpha\beta}\right)\right]\dot{\varepsilon}^{\alpha\beta} \tag{A12}$$

where

$$\begin{aligned}H = {} & 9K\frac{\partial f}{\partial I_1}\frac{\partial g}{\partial I_1} + 4GJ_2\frac{\partial f}{\partial J_2}\frac{\partial g}{\partial J_2} + 6GJ_3\left(\frac{\partial g}{\partial J_2}\frac{\partial f}{\partial J_3} + \frac{\partial f}{\partial J_2}\frac{\partial g}{\partial J_3}\right) \\ & + 2G\left(s^{\alpha m}s^{m\beta}s^{\alpha n}s^{n\beta} - \frac{4}{3}J_2^2\right)\frac{\partial f}{\partial J_3}\frac{\partial g}{\partial J_3}\end{aligned} \tag{A13}$$

Once the yield function f and the plastic potential function g are given, and the strain rate tensor as well as current stress condition are prescribed, the corresponding stress rate can be determined from Eq. (A11).

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